

Final report

about the project

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Adapted forest and fire management in climate change – Forest fire climate Resilience

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aufgrund eines Beschlusses des Deutschen Bundestages

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I. Objectives

Brief presentation of points I.1 to I.3

The project was launched to strengthen the resilience of forests and their managers to the growing challenges of climate change and the rising risk of forest fires. To this end, internationally available knowledge was to be made available to German practitioners, policymakers, and scientists. Frequent, sometimes disastrous forest fires are no longer just a southern European phenomenon; they have now reached Central Europe, including Germany. Hot, dry summers, storm damage, and beetle infestations increase vulnerability to forest fires. Therefore, adapting forests to this new risk is acute and urgent in Germany as well, requiring a strategy that goes beyond simply extinguishing fires.

The need for continuing education and knowledge sharing was consistently high throughout the project—too high to respond to all requests. WKR aimed to establish itself as a platform for knowledge transfer.

1. Task

- The scientific and technical results of the project, the achieved secondary results and the essential experiences gained, conditions under which the project was carried out
- Planning and execution of the project

The project clearly demonstrated the need for independent expertise with an outward-facing network that can act as a knowledge broker between relevant international institutions, the private sector, and the solutions market, or support training and education for those lacking the internal resources or expertise. The independent voices of WKR experts had a noticeable impact on the topic of integrated fire management nationwide, as confirmed by countless cooperation requests from other projects, government agencies, local and national working groups, and political processes, as well as by media presence and demand. By enabling multipliers from various authorities in Germany to participate in (also international) training courses and exchange of experts, WKR was able to build a high level of trust. Through such activities, institutions in Germany (e.g., state fire service schools, forestry authorities) were able to quickly expand their own capacities and competencies.

Particularly worth mentioning here is the cooperation with the industry in the field of early detection, protective clothing and forest fire hand tools, in which WKR acts in an advisory capacity for various suppliers, e.g. BOSCH, Dönges, Rosenbauer, OroraTech, Vallfirest, Mystery Ranch, Pepec, Ermak, etc. This should help to align the developments of the industry with the real needs of the users.

We now know that a timeframe of approximately three and a half years may not be sufficient to develop and refine content, and to establish nationwide networks to subsequently achieve a comprehensive, practical impact. A request for an extension could not be granted.

2. State of the art

- Scientific and technical status to which the project was linked, in particular details of known designs, processes and intellectual property rights used to carry out the project
- Indication of the specialist literature used as well as the information and Documentation services

Technical status: Valuable recommendations and fruitful collaborations have emerged from cooperation with industry in the areas of early detection, protective clothing, vehicles, and forest fire hand tools.

The networks established will continue beyond the end of the project.

z.B.:

- BOSCH: Early forest fire detection
- STIHL: Leaf blowers, multi-tools, etc. in forest fire operations
- Dönges: Compilation of the product portfolio
- Rosenbauer: Vehicle technology and cooperation with R&D Rosenauer
- OroraTech: Remote sensing, detection, monitoring and fire simulation
- Vallfirest: Equipment

- Mystery Ranch: Forest Fire Backpacks
- Pepec: Vehicle development • Lestech

Ermak: Further development of backpack sprayers, mobile water tanks and air-transportable Water tanks

This should help to align industry developments with the real needs of users.

The WKR project has been and continues to be active at trade fairs and exhibitions with many manufacturers and service providers (Interforst, Interschutz, KWF days and conferences, special shows).

3. Cooperation with other bodies

Collaborations with other WKF-funded projects and scientific institutes were also an important part of the WKR project. Collaborations and regular exchanges on progress and findings were conducted at various levels, including with the Brennpunkt Wald, PyroPhob, and THOR projects.

Furthermore, intensive collaborations and regular informal and formal exchanges with other projects, institutions, interest groups, universities, companies, service providers, forest educators, and experts were maintained as a fundamental building block of the project throughout its entire duration. These included:

- State fire brigade schools in Bavaria, Baden-Württemberg, Saxony, Brandenburg, Lower Saxony
- Forestry BW training centers Karlsruhe and Mattenhof
- Ostheide Forest Education Center
- East Bavarian Technical University of Regensburg (OTH)
- BaySF
- Bavarian Mountain Rescue Service
- THW
- Bonn Network for International Disaster Protection and Risk Management
- David Nature Conservation Foundation
- City of Erlangen
- Lower Saxony Ministry of the Interior and Sport
- Brandenburg Natural Landscapes Foundation NLB
- University of Freiburg
- Pau Costa Foundation (Spain)
- Forest Europe, Ministerial Conference for the Protection of Forests in Europe
- Forwards Project
- Nemausus Project
- Fire-In Project
- PyroLife Project
- AFAN Project
- Bernstorff's companies in Gartow
- Directorate-General for European Civil Protection and Humanitarian Aid Operations (DG-ECHO)
- UN FAO (Global Fire Management Hub)
- SNEP (Poland)
- AGIF (Portugal)
- NCC Environmental Services (South Africa)
- Further inquiries from the USA, Austria, Netherlands, Spain, Portugal, France, Czech Republic, Croatia, Belgium, Luxembourg and Lebanon.

II. Results

Detailed presentation of points II.1 to II.4

1. Results achieved

- the scientific and technical result of the project
- the achieved secondary results
- the essential experiences gained

With its products and activities, the joint project Forest Fire Climate Resilience has supported the funding policy objectives of the Forest Climate Fund in science, politics and practice.

The "Forest Fire Climate Resilience" (WKR) project has made various contributions to the funding policy objectives of the Forest Climate Fund. The Forest Climate Fund, an initiative of the Federal Ministry of Food and Agriculture (BMEL) and the Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMU), aims to preserve forests as carbon stores and strengthen their ability to adapt to climate change. Here are some of the key contributions of the "Forest Fire Climate Resilience" project to these objectives:

Prevention and risk reduction:

- **Early detection and monitoring:** The WKR project has developed technologies and methods for Developed and implemented early detection of wildfires, including remote sensing and ground-based sensors. This contributes to rapid response and containment of fires before they can cause widespread damage.
- **Risk maps and analyses:** By creating risk maps and analyzing The project was able to precisely identify areas in Baden-Württemberg that are particularly vulnerable to forest fires. This enables targeted prevention measures.

Adaptation measures:

- **Promoting resilient forest structures:** The project has exemplary measures to promote resilient forest structures, such as the diversification of tree species through natural regeneration and drone seeding and the creation of buffer zones, clearings and fire barriers. These measures improve the ability of forests to regenerate after fires according to the "Build-Back-Better" principle and reduce vulnerability to future fires.
- **Forest conversion and maintenance:** Through targeted forest management (exemplified on the demonstration areas) and the conversion of monocultures to mixed forests (WKR has circulated thousands of tree seeds and provided nursery plants), the project contributes to stabilizing the forest ecosystem and increasing resilience against climate-related stress factors.

Science and research:

- **Data and insights:** The WKR project has provided valuable insights into wildfire dynamics and their interactions with other disturbances (e.g., bark beetles, deadwood, and fire) under climate change. This information is critical for further developing wildfire mitigation and climate adaptation strategies.
- **Innovation:** Collaboration with creative partners from industry and research has resulted in innovative methods for future forest fire management under different climatic conditions. These methods assist in the planning and implementation of preventive measures.

Public relations, training and further education:

- **Awareness and education:** The project has raised awareness through professional public relations Raising awareness of the dangers of forest fires and the need for preventive measures. This includes training specialists and informing the public.
- **Networking and cooperation:** Through collaboration with various stakeholders, including forestry, fire service and research institutions, the project has promoted the exchange of knowledge and best practices.
- **Training and continuing education:** The project has created training materials for forestry and fire departments. These have now been incorporated into the first German "Forest Fire Training Standard."

Carbon storage and climate protection:

- **Forest conservation and restoration:** By contributing to the prevention of large-scale forest fires, the project helps minimize biomass loss and the release of stored carbon into the atmosphere. This supports climate mitigation efforts and contributes to long-term carbon storage.

Overall, the "Forest Fire Climate Resilience" project has made a significant contribution to the objectives of the Forest Climate Fund through its diverse approaches and measures, promoting both the prevention and control of forest fires and the adaptation of forests to climate change.

The topic of forest fires is complex, and the mandated, affected, and interested stakeholders involved exhibit a wide range of dynamics. The project's activities and results have fueled additional, independent initiatives in the broader forest fire network in a variety of ways. These secondary results can therefore never be fully described. Where possible, they will be addressed in the individual work package results.

Work Packages (WP) 1-8

Note: When submitting the application, time milestones were also defined for all planned project results.

Due to the restrictions of the coronavirus pandemic, the dependence on the availability of relevant cooperation partners, and the dynamic nature of the current fire situation during the project period, this planning became obsolete. Nevertheless, all planned results were achieved, many of them even exceeded (e.g., WP6). Constructive responses were also made to unplanned events, important requests, and issues (e.g., Forest Europe, Ukraine, areas suspected of containing munitions, etc.), thereby significantly increasing the acceptance of the project results, practical impact, and networking in WP3.

AP 1 Training modules for fire departments

The training modules for fire services were developed in close cooperation and coordination with relevant partners from the fire service and forestry sectors. The aim was to meet international standards in

to integrate the German context and ensure user acceptance of the modules. State fire schools, forestry training centers, national working groups on forest fires, and the sub-working group UAG Training and Tactics of the state-wide working group on national forest fire protection collaborated constructively in WP1. The following modules were published for free use:

Module 1: Introduction to fire behavior

The aim of this module is to introduce and describe in detail the most important elements that influence fire behavior. The following questions are answered:

- To what extent do the three elements of the so-called fire triangle (fuel, oxygen and heat energy as ignition source) influence the fire behavior?
- How do the weather factors wind, relative humidity and temperature affect the spread of fires?
- What role do propagation patterns, fire size, fire load, moisture content and type/distribution play of the fuel in the fire behavior?
- How does topography/exposure affect the spread of fires?

Module 2: Fire Weather

Aim of this module

- Influence and effect of fire weather on fire behavior.
- Effects of temperature, humidity, precipitation and wind on fire behavior • Differences between a stable and unstable atmosphere • Critical fire weather conditions

Module 3: Vegetation fire fighting: operational tactics

Objectives of the

- module • The most important strategies and tactics of vegetation fire fighting • Advantages and disadvantages of direct and indirect (defensive) fire fighting • Selection criteria for the installation and width of control lines (containment lines) • Installation of control lines on slopes

- Application and importance of operational tactics such as advance and counter fire • Detection and control of incipient fires caused by flying sparks or flying fires (spot fire)
- Effectiveness in operations, communication and teamwork • Fire watch and follow-up extinguishing work

Module 4: Vegetation fire - follow-up extinguishing work

Objectives of the module

- System and process of fire extinguishing • Extinguishing fires correctly • Finding embers • Carrying out fire extinguishing with and without water • Recognizing and minimizing dangerous situations during fire extinguishing • Fire extinguishing in difficult terrain

Module 5: Tools and Equipment

Objectives of the module

- With the correct personal protective equipment (PPE) during the use of Become familiar with hand tools • Learn about tool types and situations in which they are used • Check the usability of the hand tool, repair minor defects yourself and determine which tools need to be replaced
- Safe handling of tools in the field, especially during transport and Passing on tools • Basic information about the use of firing devices and their dangers

As a culmination of the project work, many graphics and text passages from the WKR modules were included in the First German *recommendation on training content in vegetation firefighting* by the Working Group on Natural Forest Fire Protection, UAG training adapted and adopted: <https://www.waldbrand-klima-resilienz.com/module?lightbox=datapitem-lu88bc2n>



Abbildung 1 - Vegetationsbrand bei Bismbeck 2010 ©Feuerwehrlugdienst

Empfehlung zu Ausbildungsinhalten in der Vegetationsbrandbekämpfung

UAG Ausbildung
der AG nat. Waldbrandschutz

Fassung vom 26.03.2024

1

WKR modules are also used at state fire schools in Bavaria, Baden-Württemberg, Lower Saxony, Saxony, and Brandenburg. In addition, over 100 direct requests from fire departments and forestry companies have been answered, creating a significant multiplier effect.

All modules are freely available at: <https://www.waldbrand-klima-resilienz.com/module>

WP1-relevant activities:

Date: 7-11 August 2023 **AP:** 1, 2,

7 Location:

Bad Muskau

Occasion: Training Bundeswehr Service Center **Participants:** Held,
Bundeswehr 8 people

Date: April 24, 2023 **AP:** 1,

2, 7 Location:

Online

Occasion: Training in the use of fire **Participants:** Held,
Bundeswehr Service Center

Short description: Training in preventive fire use on active military training areas

Date: April 19-21, 2023

AP: 1, 2, 3, 4, 6

Location: Erlangen

Event: Forest Fire Conference (hosted by the City of Erlangen)

Participating persons: 200 from fire service and forestry: FWF Baden-Württemberg, State Fire Service School (BW), Office for Food, Agriculture and Forestry Middle Franconia, Bavarian State Forests AöR Swabia, Bavarian State Forests AöR Upper Palatinate, Bavarian State Forests AöR Middle Franconia, Bavarian State Forests AöR Upper Franconia, Bavarian State Forests AöR Lower Franconia, Professional Fire Service Middle Franconia, Professional Fire Service Upper Bavaria, Professional Fire Service Swabia, Fraunhofer IISB, FWF Middle Franconia, FWF Lower Bavaria, FWF Upper Franconia, FWF Upper Palatinate, FWF Swabia, FWF Lower Franconia, District Fire Inspectorate Middle Franconia, District Fire Inspectorate Upper Bavaria, District Fire Inspectorate Upper Franconia, District Fire Inspectorate Upper Palatinate, District Fire Inspectorate Swabia, District Office Lower Franconia, District Office Middle Franconia, District Office Upper Franconia, District Office Swabia, private Fire service school, Lower Franconia government, State Ministry for Food, Agriculture and Forestry, Swabia city fire inspection, University of Bayreuth Chair of Biogeography, Bavarian Insurance Chamber, Lower Bavaria works fire brigade, Lower Franconia district fire inspection, Hesse volunteer fire brigade, Hesse district hazard prevention service, Mecklenburg-Western Pomerania state forest forest fire protection project THOR, Lower Saxony district fire brigade, Lower Saxony Spotfeuer / Redell Arbeitsicherheit GmbH, North Rhine-Westphalia professional fire brigade, North Rhine-Westphalia professional fire brigade Hagen, North Rhine-Westphalia volunteer fire brigade, NRW fire service institute, Rhineland-Palatinate fire service and disaster protection academy, Rhineland-Palatinate private forestry company, Saarbrücken professional fire brigade, Saxony state fire service and disaster protection school, Switzerland protection and rescue, Thuringia professional fire brigade, Thuringia Thuringia Forst AöR.

Short description: First Bavarian Forest Fire Forum

Follow up: repeated in 2024

Date: 21-13 March 2023

AP: 1, 5, 4

Location: BABZ Ahrweiler

Occasion: Working meeting of the UAG Tactics and Training of the national forest fire working group

Participants: UAG, Held and Pronto (online)

Date: March 5, 2023

AP: 1, 6, 7

Location: Beelitz Heilstätten

Occasion: Handover of WKR equipment and KBM LK Postdam Mittelmark

Participants: L. Pronto, A. Held, KBM J Heinze

Date: June 24-26, 2022

AP: 1, 5, 6, 7

Location: Elbe-Elster, Brandenburg

Reason: Major fire operation

People involved: L. Pronto

Short description: Operational support during a major fire – including firefighting and follow-up firefighting operations; on-site training of local firefighters; operational briefing with the operations command.

Discussion with Brandenburg Interior Minister Michael Stübgen, among other politicians, and with RBB-24 TV on the importance of exchanging best practices from abroad, the crucial importance of appropriate measures to contain forest fires, and the increasing need to support the training and development of more specialized forest firefighting forces.

Date: March 23-24, 2022

AP: 1, 2, 6, 7

Location: Brandenburg

Occasion: Training and demonstration area Hatzfeld Forestry Administration

Participants: L. Pronto A. Held

Date: 21 February 2022

AP: 1, 2

Location: Online

Occasion: Meeting with VWorld

Participants: L. Pronto; VWorld

Short description: Discussion on the development of e-learning tools to complement the WKR Training - especially 4-D simulator for tactical firefighting

Follow up: Will 1 pager become parameters for a 4D simulation tool for training (fighting forest fires)

Date: 21 January 2022

AP: 1

Location: Online

Occasion: Planning meeting

Participants: L. Pronto; Norman Barth (Brandenburg State Fire Service Academy)

Short description: Discussion on closer cooperation and the joint development of a training module on the use of water and equipment.

Date: September 3, 2021

AP: 1,2,3

Location: Füssen

Occasion: Working meeting with the Bavarian Mountain Rescue Service

People involved: Pronto and Held

AP 2 Training modules for forestry

WP2 was to develop training materials for forestry professionals. The modules from WP1 are equally useful as foundations for the forestry sector, as a fundamental understanding of firefighting tactics is crucial for classifying one's own forestry activities. The modules on fire behavior as a basis for planning preventive measures, as well as the fire watch and fire extinguishing module, are particularly noteworthy in the forestry context. Numerous publications (see WP6) focusing on prevention, silvicultural practices, and risk analysis have been produced specifically for WP2. The WKR partner FVA has developed a methodology for risk analysis using remote sensing data.

The FVA forest fire risk analysis was presented at numerous events nationwide, most recently at the KWF Conference 2024, special exhibition on forest fires.

The following publications for AP2 were published:

- Manual "Resilience through forest fire prevention in forestry management": https://www.waldbrand-klima-resilienz.com/files/ugd/bf6978_0d327ce13dad4dae8d814e8ea4b06382.pdf

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- Expert recommendation on silvicultural action: https://www.waldbrand-klima-resilienz.com/_files/ugd/bf6978_21f8e231b1cd47928b0ec27dd16f70d2.pdf

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In WP6 and WP7, the contents of WP2 were transferred into practice. Seminars, workshops, excursions, and training sessions at and with forestry training centers, forestry schools, and forestry and nature conservation associations and landowners sustained the transfer into forestry practice (WP2-relevant dates below). The use of the WP2 materials beyond the project term, for example, at the FBZ Karlsruhe, FBZ Mattenhof, the Brandenburg Forestry School, the German Forest Protection Association (SDW), etc., speaks to the quality and acceptance of the materials created.

WP2-relevant activities:

Date: December 5, 2023

AP: 2

Location: University of Freiburg

Occasion: Lecture on Forest Fire Management

Participants: Held, forestry students

Date: October 23, 2023

AP: 2, 7

Ort: That one

Occasion: Conference / Seminar Private Forest: Forest fire in private forest

Persons involved: Held, Forestry Chamber, SDW

Short description: Seminar with almost 80 participants

Date: October 23, 2023

AP: 2, 6

Location: Schwarzenborn, Hesse

Occasion: Entrance to KWF special exhibition on forest fires

Participants: Held, KWF, Exhibitors

Short description: Tour of the exhibition grounds for the special exhibition on forest fires organized by the WKR for the 18th KWF Conference (2024). The special exhibition will showcase the broad spectrum of interest groups that the WKR has brought together during the project period.

Follow-up: Conference in 2024

Date: September 21, 2023

AP: 2, 7

Location: Online

Occasion: SDW Online Seminar Forest Fire

People involved: Held, 150 people

Date: August 19, 2023

AP: 2, 7

Location: FBZ Karlsruhe

Occasion: Forestry BW training in forest fire basics

Participants: Held, FVA, 30 Forst BW people

Date: 11-12. May 2023

AP: 2, 7

Ort: Regensburg

Occasion: BaySF training

Participants: Held, BaySf approx. 25 people

Short description: Seminar on the basics of fire prevention: First training on forest fires for BaySF employees

Date: April 18, 2023

AP: 2, 7

Location: Bruchsal, BW

Occasion: WKR Demo Area Bruchsal: Training of 100 forestry apprentices from Forst BW

Participants: Held, FVA, 100 from Forst BW

Brief description: Planning and implementation of silvicultural measures for forest fire prevention; forest fire statistics for Germany and Baden-Württemberg (numbers and correlations); prioritization of prevention measures (map basis, decision-making basis); fire behavior (information from Module 1 WKR); young pine stands pruned; planning and implementation of silvicultural measures for forest fire prevention; equipment for initial attack by forest worker teams / experiences from operations; forestry measures from the fire service's perspective, needs and opportunities for cooperation; raising awareness of forest fire risks and dealing with forest visitors. <https://www.fva-bw.de/aktuelles/artikel/waldbrandpraevention-von-anfang-an>

Date: April 4, 2023

AP: 2, 3, 6, 7

Location: Online

Occasion: Planning and discussion of the KWF Conference 2024, special exhibition on forest fires

Participants: Held, Pronto, KWF, Exhibitors

Short description: KWF 2024, special exhibition forest fire

Follow-up: KWF conference in June 2024 including a special exhibition on forest fires

Date: 25 March 2023

AP: 2, 6

Location: Online, and in Bavaria

Occasion: ANW Bavaria, Conference on Water in the Forest (Fire Lecture)

People involved: Held

Date: March 20, 2023

AP: 2, 6, 7

Location: The Bavarian State Ministry of Food, Agriculture, Forestry and Tourism (StMELF)

Occasion: Forest Fire Conference

Participants: BaySF, STmelf, Held

Date: 15 February 2023

AP: 2, 6, 7

Location: Bingen/Online

Occasion: Planning Bingen Forest Symposium

Participants: Held, Pronto, Forestry Office Boppard RLP

Follow-up: Pronto will give a presentation on WKR successes and added value at the symposium

Date: February 7, 2023

AP: 2, 6, 7

Location: Online

Occasion: Practical workshop with "National Natural Heritage"

Participants: Held, David Foundation

Date: January 23, 2023 AP: 2,

6 Location:

Online **Event:**

Forest fire prevention Lower Saxony, State Forestry **Participants:** State

Forestry, Fire Department, EFI

Date: January 18, 2023 AP: 2,

6, 7 Location:

Online

Occasion: Discussion and planning of measures on areas of the Wildlife Foundation **Participants:** EFI, WTS,

N Semsch, Federal Forestry **Follow-up:** Measures are now being

implemented

Date: December 15, 2022 AP: 2, 6

Location:

Online

Occasion: Lecture series HAW Hamburg

Participants: A Held, students of the HAW

Date: December 8, 2022 AP: 2, 6,

5, 4 Location:

Münstertal

Occasion: Excursion with MdL R. Pix (GREEN)

Participants: R. Pix, A Held, excursion participants

Date: 3-4 December 2022 AP: 2, 5,

6 Location:

SAP Climate Arena, Sinsheim **Event:**

SAP Forest Symposium **Participants:**

A Held, approx. 30 participants

Date: November 14, 2022 AP: 2, 5

Location:

Freiburg

Occasion: Visit by a delegation from Saxony-Anhalt

Participants: A. Held **Brief**

description: The WKR project hosted a delegation of district managers from the Saxony-Anhalt State Forestry Authority to discuss "Forest Fire Management in State Forestry Operations" with WKR Project Manager Alexander Held. The colleagues traveled to Baden-Württemberg on Sunday.

Date: July 18-20, 2022 AP: 2, 5,

6 Location:

Interforst Messe München **Event:**

Interforst Messe, exhibitor with BMEL **Participants:** L.

Pronto, A. Held, M. Schloßmacher **Short description:** WKR was invited by the BMEL to present the project at their booth.

In addition to the presentation of the project, WKR experts Held and Pronto conducted numerous media interviews to raise awareness of forest fire management and to promote the concept of a holistic

To bring this approach into the public spotlight. Agriculture Minister Cem Özdemir and his Bavarian counterpart also visited the BMEL booth and learned about the various topics covered by the ministry, including forest fire prevention.

Date: April 26, 2022

AP: 2, 5, 6, 7

Location: FBZ Karlsruhe

Occasion: Forest fire training for Forest BW

Participants: A Held, FVA, Forst BW

Date: January 17, 2022

AP: 2,6

Location: University of Freiburg

Occasion: Lecture on forest fires in the Master's program

People involved: A Held, students

Short description: Lecture on fire management.

Date: January 11, 2021

AP: 2, 6

Location: University of Freiburg

Occasion: Lecture in Natural Disaster Management course, presentation of WKR project

Participants: Held, University of Freiburg Prof Hanewinkel

Short description: Lecture on Fire Management.

Follow-up: Repetition in 2022

WP 3 Exchange of Experts (EoE) and Networking

The WKR project's EoE program offered experts from forestry and firefighting, as well as other stakeholders, the opportunity to share experiences, acquire valuable knowledge, and strengthen operational skills through an exchange system. Experts could either apply to WKR for an exchange, or WKR invited specially selected candidates to suitable EoEs.

The duration of the exchange could vary from a few days to three months. The duration was typically five working days. EoEs could be conducted individually or in a small group within the WKR project. The maximum number of experts per group was 15. Participants could visit one or more organizations in one or more countries. They could, for example, observe the working structures in the visited country, learn new techniques, and study the approaches of other agencies.

They could also participate in simulation exercises as observers or attend training courses to acquire expert knowledge that is not available in Germany.

Some study and training events were designed specifically for the WKR project and, in some cases, specifically invited certain selected individuals, while in other cases WKR either organized opportunities together with partner organizations or secured participation places for WKR beneficiaries in training and exchange programs offered by other organizations.

This way, the impact was much broader, as WKR organized the participation of German actors, while many costs related to organization and logistics were borne by our partner organizations. Furthermore, such programs often involved equivalent actors from other countries, significantly expanding the networking and exchange opportunities for German actors beyond simply learning from and networking with organizations in the host country.

The WKR expert exchange program for forest fires thus made an important contribution to the further development of capacities and competencies in the field of forest risk management.

In Germany, there were also numerous theory-and-practice training courses, networking events, and expert symposiums organized directly through or supported by WKR (see WP 6 and 7). Many of these activities are listed below. The activities were often directly relevant to more than one work package, so we have organized them according to the WP focus to which the activity contributed.

Throughout the project, despite the restrictions of the Corona pandemic, expert exchanges

Exchange of Experts EoE carried out abroad:

3x EoE Poland, conducted in May 2022 and May 2023 and June 2024

- Content: Basics of fire management, interaction between forestry and fire service, Firefighting from the air, international networking
- Participants: a total of approximately 100-150 participants per EoE, of which 15-20 are from Germany

**3x EoE Portugal**

- Content: Basics of fire management, interaction between forestry and fire service, application of fire in prevention and fighting, international networking
- Participants: a total of approximately 35 participants per EoE, of which 5-10 are from Germany
- 1 long-term EoE over 4 weeks. Building future capacities for the planned Forest Fire Competence Center Brandenburg and for training at the LSTE Brandenburg



1x EoE Spain, France, Portugal

- Content: Fire study tour through 3 countries, basics of fire management, interaction of Forestry and fire service, simulation exercises, fire weather, forest fire danger index, early detection, Firefighting from the air, international networking
- Participants: 8 from Germany
- Post: <https://www.waldbrand-klima-resilienz.com/post/nemausus-spain-portugal-study-tour-and-exchange-of-experts-eoe>





2x EoE Spain (Catalonia, Gran Canaria)

- Content: Basics of fire behavior, fire analysis, basics of fire management, Interaction between forestry and fire service, fire weather, forest fire danger index, early detection, tactical fire use, use of fire for nature conservation and as a measure to reduce fuel consumption (prevention measure), international networking
- Participants: 8 from Germany per EoE

1x EoE Croatia

- Content: A specific topic of this EoE was the development of a toolbox for fire management in areas suspected of containing munitions (Unexploded Ordnance UXO). Other topics addressed included fire in difficult terrain, aerial firefighting, and a joint forest fire situation center. As the project progressed, this EoE led to demonstrations in Brandenburg and ultimately to the procurement of new technology in Brandenburg worth several million euros. The state of Saxony is now showing strong interest in pursuing the same path.
- Participants: 12 participants from Germany, forestry, fire brigade, armed forces, federal police, mountain rescue, Research
- Post: <https://www.waldbrand-klima-resilienz.com/post/exchange-of-experts-in-croatia>



1x EoE South Africa

- Content: Long-term EoE over 3 months. The content was diverse and covered the broad spectrum of integrated fire management, from community fire management to prevention and helitack.
- Participants: Two junior candidates from Germany, a trainer and multiplier at the LSTE Brandenburg, and one junior employee, to address the looming shortage of skilled workers, promote interdisciplinarity, and provide access to the international forest fire community for German participants. Both participants subsequently participated in further training and exchange programs and remain committed to the topic of forest fires in Germany.



WP3 and the EoEs conducted here have impressively demonstrated that little investment can bring a lot of knowledge, motivation, and inspiration to Germany. At the same time, this has succeeded in introducing a large number of German actors to the international forest fire network. Furthermore, as outlined in the activities listed below, there were many other exchanges, training events, international cooperation, and networking (such as the German representation

international conferences) that contributed to the objectives of this work package or even highlighted the impact on other countries.

WP3-relevant activities:

Date: December 8-9, 2023

AP: 3, 7

Location: Online

Occasion: Joint FORWARD and FoRisk Workshop

Participants: Pronto, Held, 50 participants. **Brief description:**

Experts from WKR helped organize and facilitate the Disturbance Risk Management workshop within the ForestWard Observatory to Secure Resilience of European Forests (FORWARDS) project. The workshop provided a forum for stakeholders to exchange knowledge with the goal of "evaluating disturbance risk management measures as a pillar of the CSF." The workshop was organized in collaboration with FoRISK, Forest Europe's Forest Risk Facility.

Date: November 28, 2023

AP: 3, 7

Location: Online

Occasion: Planning for the joint FORWARD and FoRisk Workshop

People involved: Pronto and Held

Date: November 22nd, 29th, 2023 **AP:**

3, 7

Location:

Online **Occasion:** Planning meeting Organizing Committee 7th Fire Behavior and Fuels Conference **Participants:**

Pronto, IAWF, Ireland, USA, Canada

Date: November 20, 2023 **AP:** 3

Location:

Online **Occasion:** Knowledge exchange with the Wildfire Safety Advisory Board of the State of California **Participants:**

L Pronto, CWSAB

Date: October 26, 2023 **AP:** 3, 7

Location:

Online

Occasion: Planning meeting for the KWF conference with the Ministry of the Interior of Lower Saxony

Participants: Held, Pronto, NI IM

Date: 19-20 October 2023 **AP:** 3, 7

Location:

Online

Occasion: Planning for the joint FORWARD and FoRisk Workshop **Participants:**

Pronto and Held

Date: October 10, 2023

AP: 3, 7

Location: Online

Occasion: Northwest European Roundtable on Wildfire Management

Participants: Pronto, Held, participants from UK, NL, PL, DE, ES, PT, FR

Short description: regular exchange with forest fire stakeholders

Date: September 14-17, 2023 **AP:** 3, 7

Location:

Czech Republic

Event: Czech Wildfires Academy **Participants:**

Pronto, PCF, 150 firefighters **Brief description:** Pronto acted as a

trainer in a training camp inspired by the WKR-SNEP Forest Camp model. WKR training materials were distributed to the participants (mainly Czech, but also senior officers of the GFFF-EU module from Slovakia). Promoting consistent best practices is important in cross-border fires and EU operations.

Follow up: Czech Wildfire Academy 2024

Date: August 9, 2023 **AP:** 3, 7

Location:

Online

Occasion: Planning meeting Organizing Committee 7th Fire Behavior and Fuels Conference **Participants:** Pronto,

IAWF, Ireland, USA, Canada

Date: 31 July 2023 **AP:** 3

Location:

Online **Occasion:** Working meeting on the technical recommendation "Wildland Urban Interface"

Participants: L. Pronto, OTH, East Allgäu

Date: 19 July 2023 **AP:** 3

Location:

Online **Occasion:** Planning for the technical recommendation "Wildland Urban Interface"

Participants: L. Pronto, OTH, East Allgäu

Brief description: Lindon will prepare a first draft based on basic literature and international best practices. OTH and OA will write a first draft.

Date: 25-29. May 2023

Location: Gubin, Poland

AP: 3, 1, 2

Occasion: EoE Poland (SNEP Forest Camp)

Participants: L. Pronto, A. Held, many WKR partners, 150 participants from 22 countries, including 15 from Germany

Short description: Expert exchange to Portugal. WKR was able to enable German forestry and firefighters to participate in a week-long training camp in Portugal. The technical content was developed with extensive international participation. This EoE therefore not only provided advanced training but also expanded networking in the international forest fire sector for German participants. Blog post: <https://www.waldbrand-klima-resilienz.com/post/fuelling-the-policy-from-groundwork-to-an-international-audience>

Follow up: Forest Camp 2024

Date: 7-11. May 2023

AP: 3

Ort: Mafra, Portugal

Occasion: EoE Portugal

Participants: 6 from Germany

Short description: International exchange on fire prevention and the use of fire; WKR was able to enable six German forestry and firefighters to participate in a week-long training camp in Portugal. The technical content was developed with extensive international participation. This EoE therefore not only provided advanced training but also expanded networking in the international forest fire field for German participants.

Follow up: annual event

Date: April 27-28, 2023

AP: 3, 6

Location: Wildau, Brandenburg

Occasion: Network meeting Civil Protection Brandenburg

Persons involved: Held, fire departments, aid organizations, municipal administrations, district administrations, city administrations, state administrations, forestry, armed forces, clinics, rescue services, control centers, federal police, companies

Short description: <https://www.netzwerk-bevoelkerungsschutz.de/3.-netzwerktreffen/>

Date: April 17, 2023

AP: 3, 7

Location: Bingen

Occasion: Forest Symposium Bingen on the topic of forest fire management

Participants: Pronto, 200+ participants in person and 600+ online

Short description: <https://www.wald.rlp.de/mediathek/wald-veranstaltungen> WKR was presented to an interested audience and this well-attended symposium resulted in numerous collaborations.

Date: April 12, 2023

AP: 3, 7

Location: Online

Occasion: Northwest European Roundtable on Wildfire Management

Participants: Pronto, Held, participants from UK, NL, PL, DE, ES, PT, FR

Short description: regular exchange with forest fire stakeholders

Follow up: 2x annually

Date: December 6, 2022

AP: 3

Location: Online

Occasion: Meeting with NASA (USA) and University of Maryland, Baltimore County

Participants: L. Pronto, A. Held

Short description: Discussion about potential synergies and possible activities for knowledge transfer from the NASA program "Earth Science Applied Sciences Wildfires" to Germany.

Date: November 15-16, 2022

AP: 3, 5, 6

Ort: Larnaca, Cyprus

Anlass: 4th Nicosia Risk Forum: "Disaster Management Challenges and Solutions in the Era of Climate Crisis" hosted by European University of Cyprus (CERIDES)

People involved: L. Pronto

Short description: Lindon moderated a session on six European projects and initiatives related to wildfires and presented the WKR project as a pan-European best practice at the 4th Nicosia Risk Forum. The conference was opened by the European Commissioner for Civil Protection and Humanitarian Aid, and over 200 leading civil protection agencies from Europe, the Middle East, the US, and Mexico participated physically and virtually. The forum was preceded and followed by exploratory meetings with Ambassador Grech, Under-Secretary-General of the Union for the Mediterranean, and the French Civil Security Official for European Affairs and Civil Protection (for Civil Protection) seconded by the UfM.

Date: November 8-10, 2022

AP: 3, 6

Ort: Belfast, Ireland

Anlass: UK Wildfire Conference, Northwest EU roundtable workshop

Persons involved: A. Held

Date: 15-21 October 2022

AP: 3

Location: Gran Canaria, Spain

Occasion: Exchange of Experts (Spain)

Participants: L. Pronto, and WKR partners: Mountain Rescue Bavaria, LFS Saxony, Forest Fire Team, @fire, SNEP, PFC

Lindon worked as an operations manager/group leader and instructor in Gran Canaria and, in cooperation with PCF, taught 20 specialists as part of an international exchange and the

Capacity building, among other things, for beneficiaries of the WKR project in the use of fire. The one-week course included one day of classroom instruction, three full days of firefighting, and a field trip. Firefighting was conducted using the Incident Command System and supported by approximately 50 Spanish firefighters and members of the civil protection agency. Through WKR, Bosch was also invited to test its AI and sensor technology for early fire detection in a realistic environment. The training was accompanied by a documentary film crew from the German television program Plan B (ZDF).

Date: 10-17 September 2022

AP: 3

Location: Croatia – Zagreb, Zadar, Split, Rijeka, Duvilje

Occasion: EoE Croatia

Participants: L. Pronto, A. Held

Short description: <https://www.waldbrand-klima-resilienz.com/post/exchange-of-experts-in-croatia> and Croatian press release: <https://hvv.gov.hr/vijesti/posjet-njemackih-vatrogasaca/3303>

Date: August 2, 2022

AP: 3

Location: Online

Occasion: Planning meeting for EoE Croatia

Participants: L. Pronto, A. Held, K. Filcic

Date: July 18, 2022

AP: 3

Location: Online

Occasion: Working meeting with the Forest Fire Working Group of the Bonn Network for International Disaster Management

People involved: L. Pronto

Date: July 8, 2022

AP: 3, 6

Location: Online, Ukraine, Germany

Occasion: Zoological Society Frankfurt Meeting on aid fire in Ukraine

Participants: L. Pronto, A. Held, FZS

Short description: FZS areas in Ukraine have fire problems

Follow up: Equipment assembled and delivered

Date: June 27-30, 2022

AP: 3

Location: Brussels, Belgium

Occasion: 7th European Civil Protection Forum

People involved: L. Pronto

Short description: Participation as a German expert; side meetings with DG-ECHO, BBK, THW and others; networking and active contributions in various workshops; contributions to challenges for German emergency services

Date: June 14, 2022

AP: 3, 5, 6

Location: Online

Occasion: Working meeting with BSOCH Sensortechnik for early forest fire detection

Participants: L. Pronto, A. Held

Date: May 24-27, 2022

AP: 3, 6

Location: Pasadena, CA, USA

Anlass: International Association of Wildland Fire Fire and Climate Conference

People involved: Lindon

Short description: Lindon Keynote as internationally nominated "Emerging Leader" in Forest fire management

Date: May 19-22, 2022

AP: 3

Location: Golabski, Poland

Occasion: Exchange of Experts Poland

Participants: Lindon, Alex, Maria and about 100 other participants from 8 countries, including Craig and Ciaran as instructors

Short description: SNEP-WKR EoE – see blog post

Date: 11. May 2022.

AP: 3, 7

Location: Online

Occasion: Planning meeting with the Füssen Mountain Rescue Service

Participants: L. Pronto, BW-B, OTH

Date: April 24-29, 2022

AP: 3

Location: Barcelona, El Bruc, Madrid (Spain), Lisbon (Portugal)

Occasion: Expert exchange Spain/Portugal **Participants:**

Ministry of the Interior (Croatia), Ministry of the Interior (France), Métropole Nîmes (France), Hellenic Fire Brigade Corps (Greece), Pau Costa Foundation (Spain), Special Unit for Civil Protection (GAUF-FEPC) (Portugal), Association of Independent Fire Experts (SNEP) (Poland), DREAM Italia (Italy), European Forest Institute (Germany), East Bavarian Technical University of Regensburg (Germany), Bavarian State Fire Service School, Regensburg (Germany), Brandenburg State Academy for Fire and Civil Protection (LSTE) (Germany), Lower Saxony State Office for Fire and Civil Protection (NLBK) (Germany), Forest Fire Team (NGO) (Germany).

Short description: A one-week study tour in Spain and Portugal, organized by the Forest Fire Climate Resilience (WKR) project in collaboration with partner PCF. This tour was part of a collaboration between the AFAN project and the NEMAUSUS project of the European Union Civil Protection Mechanism. During this tour, Spanish and Portuguese colleagues shared their knowledge and research with fire management experts from Germany, with the aim of improving understanding and methods in fire management.

Date: April 19-22, 2022

AP: 3, 7

Location: Marseille, France

Occasion: Expert exchange and workshop – Advanced Fire Analysis Network (AFAN)

Participants: L. Pronto, OTH (WKR partner); PCF, SDIS-13, European forest fire experts and civil protection authorities (from 12 countries)

Short description: AFAN information event and workshop; study tour to the French fire service school in Vallabre and site visit, as well as field exercises with local fire services and municipalities; Pronto delivered a keynote speech and participated in a panel discussion on the challenges facing European firefighters, including examples from Germany and initiatives such as WKR.

Date: April 15, 2022

AP: 3

Location: Online

Occasion: Training offers for the farmers of the machine rings

Participants: Alex, Lindon, WaldbrandTeam and Machinenring Bayern

Short description: Discussion on how to train farmers in fire prevention and first response skills.

Follow-up: Contribution to a catalogue of various agricultural equipment and their possible Use to support firefighting

Date: March 1-4, 2022

AP: 3

Location: Bardo, Poland

Occasion: Working meeting to prepare EoE Poland

Participants: A Held, SNEP

Date: 11 February 2022

AP: 3

Location: Online

Occasion: NASA Science Mission Directorate (SMD) Wildfire Stakeholder Engagement Workshop.

Participants: L. Pronto; NASA, USDA, USGS + 600+ international participants **Brief description:** Stakeholder workshop breakout sessions; discussion and contributions in Breakout 2: Active fire (detection & tracking) and *Breakout 4: Pre-fire (fuels/ignitions)*

Follow up: Publication

Date: 03-04 February 2022

AP: 3

Location: Online

Occasion: NASA Science Mission Directorate (SMD) Wildfire Stakeholder Engagement Workshop

Participants: 600+ authorities, interest groups, NGOs and organizations

Brief description. Workshop and exchange on the different needs of countries in the area of space-based information on forest fires. WKR represents the needs of Germany, discussions on cooperation with the European Commission, and the deficiencies of nationally available space-based information on forest fires for the German context.

Date: 26 January 2022

AP: 3, 6

Location: Online

Occasion: Information meeting

Participants: L. Pronto; WKR partner Pau Costa Foundation; Research Department Rosenbauer

Short description: Discussion about innovation with reference to challenges/opportunities/needs in Germany, Austria, Europe

Date: 24 January 2022

AP: 3

Location: Online

Occasion: Bonn Meeting Bonn Network for International Disaster Management

Participants: WKR, DKKV, FW /City of Bonn, Fraunhofer, DHL, UNU, University of Bonn, Welthungerhilfe, City of Cologne, Germany Helps.

Short description. Preparation of the webinar: Forest fire activities in Germany and Europe; introduction of WKR

Follow up: Webinar March 5, 4:00 PM

Date: 07 December 2021 AP: 3

Location:

Online **Occasion:** Cooperation, expert exchange, training, cross-border operations **Participants:** L. Pronto; IFV (Brandweeracademie /Institute for Safety) Netherlands **Short description:** Follow-up activity to the establishment of Northwestern European Fire Management Network (control meeting); €100,000 received for 2 years

Date: 10 November 2021 AP: 3, 6

Location:

Online **Event:**

UK Wildfire Webinar **Participants:** L.

Pronto; 160 participants from 18 countries **Short description:** Keynote on improving cross-border firefighting cooperation in Europe

Date: 24-30 October 2021 AP: 3

Location: Catalonia, Spain

Occasion: Training on Technical Fire Use /exchange on training development **Participants:** L.

Pronto **Short description:** One-week

technical course + preparation for the development of a technical course for the use of fire in Germany (preventive and as a firefighting measure)

Date: 13 October 2021 AP: 3

Location:

Online **Event:** Northwestern European Roundtable on Fire Management **Participants:**

L. Pronto **Short description:** Keynote on cooperation in Europe

Date: 07 October 2021 AP: 3, 8

Location:

Online

Occasion: Cooperation with WKR Partner PCF (Spain)

Participants: L. Pronto; PCF **Short description:**

Discussion of the cooperation and synergies between the WKR training modules and the PCF training materials

Date: September 15, 2021 AP: 3

Location:

Online **Event:** Webinar: Envisioning Different Firefighting Futures. Futures Assessed Alongside Socio-technical Evolutions **Participants:** L. Pronto, A.

Held; Fire-IN consortium; FAO + international wildfire experts **Brief description:** Webinar and workshop (Pronto moderated)

Date: August 23, 2021 AP: 3, 6

Location:

Online

Occasion: Meeting with OroraTech

Participants: Held, Pronto, Stoffers **Short description:**

Introduction to OroraTech's Online Wildfire Platform **Follow up:** Global user license for Held and Pronto

Date: August 23, 2021 AP: 3

Location:

Online **Occasion:** Bonn Meeting of the Bonn Network for International Disaster Management **Participants:**

WKR, DKKV, FW /City of Bonn, Fraunhofer, DHL, UNU, University of Bonn, Welthungerhilfe, City of Cologne, Germany Helps.

Brief description. Exchange and planning

Date: August 23, 2021

AP: 3, 6

Location:

Online **Occasion:** Instructions on Ororatech Software Operation

Participants: A. Held, L. Pronto; Ororatech **Short description:**

Tutorial on the use and possibilities of Ororatech's real-time assessment and monitoring of forest fires

Date: June 29, 2021 AP: 3

Location:

Online **Occasion:** Bonn Meeting of the Bonn Network for International Disaster Management **Participants:**

WKR, DKKV, FW /City of Bonn, Fraunhofer, DHL, UNU, University of Bonn, Welthungerhilfe, City of Cologne, Germany Helps.

Brief description. Exchange and planning

Date: June 22, 2021 AP:

3, 6

Location:

Online **Occasion:** Consulting a German start-up to support meaningful development for practitioners.

Participants: WKR (Pronto) and Dryad Networks **Short description:**

Discussion about the market demand for early detection of forest fires in Germany

Date: May 28, 2021 AP: 3,

6 Location:

Online

Occasion: Meeting of the Bonn Network for International Disaster Management **Participants:**

WKR, DKKV, FW /City of Bonn, Fraunhofer, DHL, UNU, University of Bonn, Welthungerhilfe, City of Cologne, Germany Helps.

Brief description: Description of network activities, skills building, and synergy sharing. Discussion on the establishment of working groups.

Follow up: Establishment of the Forest Fire Working Group within the Bonn Network for International Disaster Management

AP: 3, 6

Ort: Online

Anlass: Northwest European Round Table on Wildfire **Beteiligte**

Personen: Ca. 35 ppl from 8 countries and 20 organizations **Kurze Beschreibung:**

Exchanges, presentations and discussions on conditions, experiences and initiatives in NW European Countries to learn from one another.

Participation from 8 countries. Germany represented by WKR Team, Waldbrand Team, FW Bonn, and Fraunhofer **Follow up:** next round table in 6 months

Datum: 12-16. April 2021 AP: 3

Ort:

Placer and Nevada County California, USA **Anlass:** Prescribed

Fire TRaining EXchange "Cal-TREX Yuba"

Participants: Pronto + 20 stakeholders from Northern California **Short description:**

Prescribed Fire Training Exchange "Cal-TREX Yuba" hosted by Open Canopy LLC and the Yuba Watershed Institute supported by The Nature Conservancy and CALFIRE

Date: January 18, 2021 AP: 3,

6 Location:

Online

Occasion: Meeting with Rosenbaur International re research initiative forest fire **Participants:** Held, Rosenbaur

Short description Introduction of WKR at

Rosenbaur research department and planning of Rosenbaur research on the topic of vegetation fire 2030.

Follow up: Herbst 2021.

Date: December 7, 2020 AP: 3

Location:

Online **Occasion:** Meeting with Maschinenring Deutschland, new cooperation partner **Participants:** Süssner,

Held, Leisgang **Brief description:** Maschinenring DE has

290 locations in Germany with equipment and machinery that can play a role in the prevention, suppression, and aftercare of forest and vegetation fires. Exchange of ideas on cataloging the equipment in Germany and the activation mechanism.

Follow up: Article at Kohlhammer 2021 and further planning and description of the requirements for agricultural equipment.

Date: November 5, 2020

AP: 3, 7

Location: Online

Anlass: TREX planning for California

Participants: L. Pronto, S. Graydon

Kurze Beschreibung: *We got funded by The Nature Conservancy for a 1-week training!* Discussion on the planning process for organizing TREX training event in California **Follow up:** Check in in Feb/March (link to organizing folder where planning is ongoing).

Date: November 2, 2020

AP: 3

Location: Online

Occasion: Planning meeting with the U.S. Forest Service Washington DC Office and International Programs Eastern Europe/Caucasus

Participants: A. Held, L. Pronto Wills, Matthew; Slemph, Shelia; Murphy, Tim

Kurze Beschreibung: Discussions on introducing fireline explosives in Germany for UXO terrain and Hazard Trees (in conjunction with THW); training requirement for being USFS recognized trainer in DE; interoperability in Europe

Date: 19 October 2020

AP: 3, 7

Location: Online

Reason: TREX planning for California (due to transfer for Germany)

Participants: L. Pronto, S. Graydon

Kurze Beschreibung: Discussion on the planning process for organizing TREX training event in California **Follow up:** Wait for notification of funding approval

Date: September 22, 2020

AP: 3

Location: EFI Bonn

Occasion: Municipal Agency NRW

Participants: Alex, Andreas, (Marcus), Maria (EFI)

Short description: Municipal agency offers services for municipalities. Interested in addressing the topic of WKR in municipalities. Awareness work and implementation of preventive measures. E.g. Information day next year, e.g. April

Follow-up: Which topics can be covered with which speakers? Alex prepares an email

Date: August 21, 2020

AP: 3

Location: Online

Reason: Planning meeting with the US Forest Service Washington DC Office and International Programs Eastern Europe/Caucasus

Participants: A. Held,

L. Pronto Wills, Matthew; Slemph, Shelia; Murphy, Tim **Follow-up** cooperation with the USFS, loose but coordinated: Design and content of the training modules in the style (Series) of Congo, as the start of a series.

WP 4 Development of the framework and core content of a general forest fire strategy

A cross-border, nationwide forest fire strategy or long-term vision would be extremely important in providing reference, leadership and orientation, a framework, and political support on the issue of forest fires in a climate of change. Furthermore, a national strategy could be helpful in the context of European cooperation and coordination. For example, *Forest Europe* has defined forest fires as a priority of the European Forest Risk Facility (based in Germany) under Forest Europe. From a project perspective, however, there was little interest at the federal level in addressing the issue of a national forest fire strategy, as forest fire management is a federal state responsibility. Various federal states are now developing strategies, concepts, and forest fire competence centers without a comprehensive framework.

Under these conditions, the WKR project has participated in various state working groups and processes. The partner FVA was and remains involved in the **Federal-State Working Group on Forest Fires (BLAG)**, which, however, has not shown any signs of developing a genuine national strategy.

Between 2020 and 2022, the WKR project was invited to participate in various processes/phases to define the future requirements for forest fire management in the Free State of Bavaria. The process was conducted under the auspices of the Ministry of the Interior and implemented by the Regensburg State Fire Service School.

and the OTH University

Report here: <https://publikationen.bibliothek.kit.edu/1000150496>

In 2022, WKR was invited to join the **Lower Saxony Forest Fire Expert Commission**.

The WKR project had a particularly strong impact, as it was the only non-emergency services organization to be included in the **UAG training program of the state-wide Working Group on National Forest Fire Protection** (Working Group V of the Standing Conference of the Interior Ministers and Senators of the States (AK V of the Conference of Interior Ministers)). After three years of collaboration, the UAG training program issued its recommendations to the Working Group on National Forest Fire Protection in 2024 to establish harmonization of training and tactics in national vegetation firefighting - a large portion of the graphics and training content were taken directly from and adapted to the WKR training modules.

All modules from AP1 were analyzed in detail in the UAG training workshops, adapted, and largely incorporated into the training standard, both in terms of content and graphics.

In addition, at the request of the working group, WKR developed new original graphics to complement the recommendation with visually consistent graphics that illustrate concepts not covered in WKR Modules 1-5 under WP1. In the summer of 2024, the graphic designer responsible for the WKR module graphics and layout will create the final version of the UAG recommendation, delivering a stylistically and visually consistent result with the WKR WP1 deliverables. Since the working group process has been ongoing for three years, many volunteer fire departments, cities, and counties have initiated their own internal processes to either review current training concepts and disaster management plans or develop new training programs to better prepare for the challenge of wildfires. Subsequently, the WKR project has reached an agreement with the Lower Saxony Ministry of the Interior (Chair of the UAG Training) to distribute the recommendation, which is not yet publicly available, to relevant stakeholders upon request to avoid parallel efforts in pursuing a national training standard. We have distributed the recommendation to the following organizations upon individual request until August 2024:

- Wöhratal Volunteer Fire Department
- Bamberg district
- Volunteer Fire Department Schliersee / Upper Bavaria
- Wittmund district / East Frisia
- Bavarian State Institute for Forestry • Oberursel Volunteer Fire Department
- City of Kassel
- Volunteer Fire Department Böhl-Iggelheim
- Vegetation fire training group Saarlouis district
- Hambühren local fire department (Celle district)
- Bundeswehr Fire Department Munster
- Volunteer Fire Department Kirchehrenbach / Bavaria
- Volunteer Fire Department City of Naumburg / Hesse
- Adelebsen Community Fire Department
- Uckermark district / Brandenburg
- NIPV (Netherlands Institute for Public Safety)
- Volunteer Fire Department Geilenkirchen / NRW
- Volunteer Fire Department Großbeeren OT Kleinbeeren
- Volunteer Fire Department Florstadt Nieder-Mockstadt
- Volunteer Fire Department Ballmertshofen / Blauwald GmbH (BW)

- Government of Lower Bavaria
- Kahla Volunteer Fire Department
- Volunteer Fire Department Holzschlag (Austria)
- Volunteer Fire Department Höchst i. Odenwald – West.
- Kiel City Fire Department / Schleswig-Holstein State Fire Department School Holstein
- Volunteer Fire Department Altdorf (Ostallgäu, BY)
- Volunteer Fire Department Heiligenhaus • City of Mannheim Fire Department & Disaster Control
- Volunteer Fire Department Schlagsdorf (MV)
- Darmstadt Fire Department

Empfehlung zu Ausbildungsinhalten in der Vegetationsbrandbekämpfung

Für Personen, die in ihrer Freiwilligen Feuerwehr, ihrer Stadt oder ihrem Landkreis mit der Entwicklung / Durchführung von Schulungen zum Thema Waldbrand beauftragt sind, gibt es eine neue Empfehlung der Arbeitsgruppe nat. Waldbrandschutz, UAG Ausbildung. Die aktuelle Fassung wird an die Innenministerkonferenz weitergeleitet mit der Absicht, sie an die Länder mit der Bitte um Anwendung bzw. Einführung zu verteilen. Um parallele Bemühungen zu vermeiden, haben wir die Erlaubnis, dieses Dokument auf schriftliche Anfrage an Sie weiterzugeben.

Wir sind sehr dankbar, dass das WKR-Projekt das Privileg hatte, während dieser mehrjährigen Zusammenarbeit in der UAG, wichtige Beiträge zu liefern!



Abbildung 1 - Vegetationsbrand bei Barmbeck 2010 (Feuerwehrjugendrat)

Empfehlung zu Ausbildungsinhalten in der Vegetationsbrandbekämpfung

UAG Ausbildung
der AG nat. Waldbrandschutz

Fassung vom 26.03.2024

WKR was and is an active member of the **AK Waldbrand of the German Fire Brigade Association**.

The development of core elements of a national forest fire strategy cannot be achieved without coordination with European neighbors and global forest fire initiatives and processes. WKR has therefore also actively participated in this process and ensured coordination with international processes. The various initiatives are briefly presented below:

The recommendations of the European Commission (**Sparking FireSmart Policies for the EU**) were adapted for the German context and published in the form of various presentations and a term paper.



Landscape Fire Governance Framework (LFGF). WKR was able to participate in the development of the LFGV and, as the German participant, present the LFGF at the **International Wildfire Conference (IWFC) 2023** in Portugal.

As a result, WKR, together with the Global Fire Monitoring Center GFMC, was the only German initiative to introduce and publicize the LFGF in Germany.

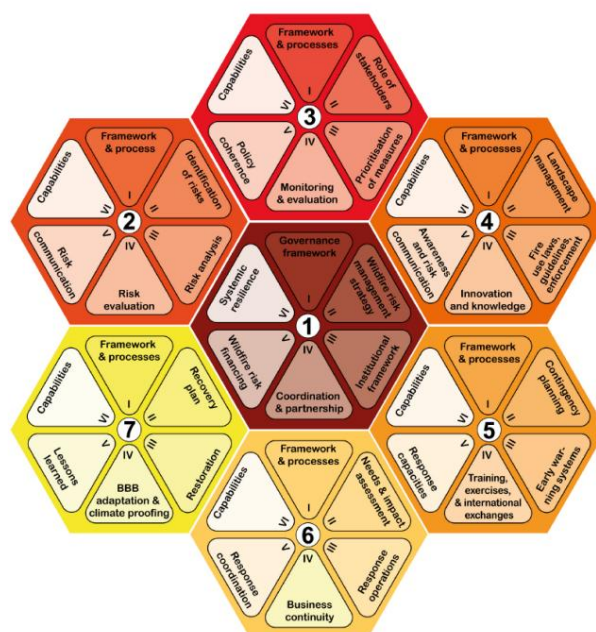
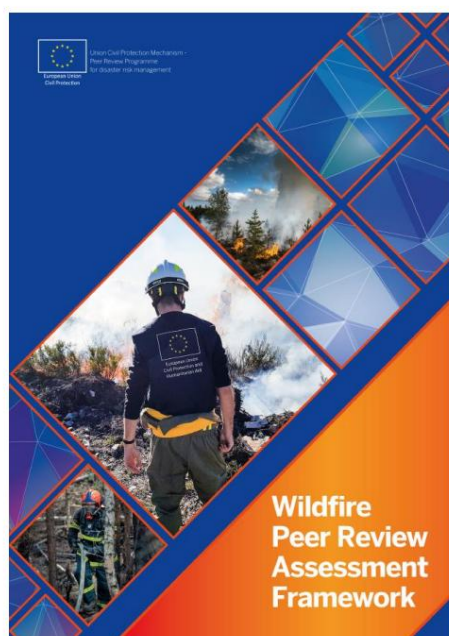


Wildfire Peer Review Assessment Framework (Wildfire PRAF). WKR experts co-developed the Wildfire PRAF for **DG ECHO** of the European Commission. The Wildfire PRAF is used to evaluate a country's forest belt situation and forms the basis for uniform European standards and methods in fire management, a basic prerequisite for an internationally coordinated forest fire strategy. **Through the networking of the WKR project, the Wildfire PRAF will now be applied in Brandenburg and Saxony. In Brandenburg, the results will serve as a basis for**

Defining the scope of action and the range of tasks of the planned Forest Fire Competence Center. However, there appears to be no interest at the federal level in implementing this for the entire country.

Within the framework of the Wildfire PRAF, the following content components for a functioning forest fire concept or strategy are analyzed and evaluated, and subsequently recommendations are formulated for improvement and to achieve comparability at the European level.

- Introduction
- Overview of the thematic areas of the wildfire peer review
 - o Governance of wildfire risk management
 - o Wildfire risk assessment
 - o Wildfire risk management planning
 - o Wildfire prevention
 - o Wildfire preparedness
 - o Response
 - o Recovery and lessons learned
- Annexes



Study for the Committee on Regional Development of the European Parliament

WKR experts led and contributed to a study for the European Parliament's Committee on Regional Development, which assessed the 2022 European forest fire season, the European response to these fires, and the impact of various European funding mechanisms for fire prevention, preparation, response, and management. Numerous specific challenges in the German context were highlighted (e.g., the munitions-contaminated terrain), and WKR was mentioned as a best practice for developing innovative solutions. Numerous German stakeholders were interviewed, and relevant passages of the report can be found, for example, on pages 12, 16, 20, and 47-48 of the report.

German summary: [https://](https://www.europarl.europa.eu/RegData/etudes/STUD/2023/747280/IPOL_STU(2023)747280(SUM01)_DE.pdf)

[www.europarl.europa.eu/RegData/etudes/STUD/2023/747280/IPOL_STU\(2023\)747280\(SUM01\)_DE.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2023/747280/IPOL_STU(2023)747280(SUM01)_DE.pdf)

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Study: [https://www.europarl.europa.eu/thinktank/en/document/IPOL_STU\(2023\)747280](https://www.europarl.europa.eu/thinktank/en/document/IPOL_STU(2023)747280)

ZUSAMMENFASSUNG



STUDY
Requested by the REGI Committee



Forschung für den REGI-Ausschuss – Waldbrände im Sommer 2022

Lehren aus der Kohäsionspolitik



Das Jahr 2022 verzeichnete einen deutlichen Anstieg der Waldbrände in ganz Europa, wobei der Schwerpunkt auf Nationen wie Frankreich, Spanien, Rumänien, Deutschland, Tschechien und Slowenien lag. In einigen Fällen war die verbrannte Fläche 5-13 Mal höher als im Durchschnitt der letzten zehn Jahre und hat sich auf eine Gesamtfläche angesammelt, die mehr als das Dreifache der Größe Luxemburgs beträgt. Dieser Anstieg der Waldbrände wurde durch anhaltende Hitzewellen, Dürren, besonders im Frühjahr, und starke oder ungewöhnliche Windmuster verschärft. Trockene Bedingungen führten zu den niedrigsten registrierten Bodenfeuchte seit fünfzig Jahren und 63 % der Flüsse sind weit unterdurchschnittlich abgeflusst, was zeigt, dass 2022 das trockenste Jahr in der jüngeren Geschichte war. Einige Indikatoren für diese Bedingungen waren eine erhöhte Brandaktivität im Alpenraum und eine gesteigerte Häufigkeit von extremem Brandverhalten sowie Pyrocumulonimbus-Formationen (noch reichende Gewitterwolken, die durch Brände entstehen) im Mittelmeerraum. Länder und Regionen, die traditionell als geringes Risiko für Waldbrände gelten, sahen sich mit großflächigen Bränden und extremem Feuerverhalten konfrontiert.

Der Mangel an Erfahrung, Vorsorge und angemessenen Ressourcen in diesen Bereichen hat die Eindämmungsbemühungen erheblich behindert. Darüber hinaus hat sich ein wirksames Waldbrandmanagement in Regionen mit nicht detonierten Kampfmitteln (auch Blindgänger genannt) (UKO) als entscheidendes Anliegen herausgestellt, insbesondere angesichts der Vorfälle in Slowenien und Deutschland sowie der erhöhten Risiken in Konfliktgebieten wie der Ukraine. Ukraine war 2022 das am zweithäufigsten von Bränden betroffene Land Europas. Europas Schutzgebiete, besonders die Naturazoo-Gebiete, berichteten ebenfalls von einem Anstieg von Waldbränden und verbrannten Gebieten, was auf ein kritisches Fehlen umfassender Brandbewältigungsstrategien hindeutet.

Das vorliegende Dokument ist die Zusammenfassung der Studie über Waldbrände im Sommer 2022. Die vollständige Studie, die in englischer Sprache verfügbar ist, kann heruntergeladen werden unter: <https://bit.ly/4zv6oKI>

Fachabteilung Struktur- und Kohäsionspolitik
Generaldirektion Interne Politikbereiche
Autor: Lindon PRONTO (Stiftung Pau Costa)



Policy Department for Structural and Cohesion Policies
Directorate-General for Internal Policies
PE 747.280 - December 2023

EN

Forest Fires of Summer 2022

Lessons to Draw from the Cohesion Policy Response



Regional Development

UNEPI/FAO Global Fire Management Hub. Experts from the WKR Project provided input to define the priorities and scope of the Fire Hub through a series of technical workshops between 2023 and 2024. The Fire Hub brings together key partners to support countries in building capacity to implement integrated fire management to reduce the negative impacts of wildfires on livelihoods, landscapes, and global climate stability. With initial funding of US\$5 million from the Canadian Forest Service, the German Federal Ministry of Food and Agriculture, the Korea Forest Service, the Portuguese Agency for Integrated Rural Fire Management, the US Forest Service, and USAID1, the Fire Hub addresses the following five areas:

1. Improving knowledge exchange by defining key firefighting concepts, developing new publications and guidelines, and conducting workshops and events to improve the availability of technical advice on integrated fire management;
2. Strengthening capacity building through training with national actors and Mentoring program for young fire experts;
3. Improving methods and tools for fire risk assessment and early warning, as well as the associated forest fire risk assessment;
4. Creating equitable and diverse communities resilient to wildfires, collaborating with indigenous peoples and building on traditional knowledge in a community-based approach to fire management; and
5. Providing political support to help governments better implement integrated fire management can be integrated into national and subnational policies.²

WP4-relevant activities:

Date: 9. November 2023

AP: 4, 5

Location: Online

Occasion: Regular meeting of the Forest Fire Expert Commission, State of Lower Saxony

People involved: L Pronto, A Held

Date: September 19-21, 2023

¹ <https://www.fao.org/newsroom/detail/faolaunch-updated-guidelines-to-tackle-extremewildfires/en>

² <https://www.fao.org/forestry-fao/firemanagement/101248/en/>

AP: 4, 5

Location: Celle, Lower Saxony

Occasion: Working meeting of the UAG Tactics and Training of the national forest fire working group

Participants: UAG, Pronto **Short**

description: Working meeting with excursion to the military training area to inspect the new Tatra forest fire vehicle + procurement; visit to the German Air Tractor Squadron of the RescEU (UCPM) in Braunschweig and meeting with pilots.

Date: June 30, 2023

AP: 4, 5

Location: Online

Anlass: Expert Working Group of FoRisk (Forest Europe)

Participants: Held, Pronto, BMEL, FE, 15-20 countries

Short description: Planning for the Forest Risk Facility (hosted in Germany) and review of the results of the Wildfire Pilot project led by WKR

Date: May 15-20, 2023

AP: 4, 5, 3

Ort: Porto, Portugal

Occasion: 8th International Wildland Fire Conference, WKR represents DE

Participants: Held, Pronto, Schloßmacher, WKR partners, 1200 participants

Brief description: WKR participated as Germany's representative at the International Forest Fire Conference. Many colleagues from the international forest fire network were reunited. WKR organized side events and actively participated in various workshops. We are particularly proud of WKR's participation in the [Landscape Fire Governance Framework](#).

Date: April 17, 2023

AP: 4, 5

Location: Berlin

Occasion: Hearing before the **Interior Committee of the German Bundestag**

People involved: Held

Short description: Expert hearing on the topic of forest fires in Germany:

The Bundestag's Interior Committee is holding a public hearing on the topic of forest fire protection.

The experts invited by the parliamentary groups called for improvements in forest firefighting. Blog post and further

documents: <https://www.waldbrand-klima-resilienz.com/post/experten-fordern-bessere-waldbrandbek%C3%A4mpfung>

Date: February 3, 2023

AP: 4, 6

Location: Berlin

Occasion: Parliamentary Breakfast: Lecture on Forest Fire

Persons involved: Parliamentarians' staff, SDW, EFI

Short description: WKR project manager Alex Held was invited to give a lecture on forest management as part of their "Forest Knowledge" event series on February 3 in Berlin. This was the second event in this format and is aimed at members, speakers, and staff of the German Bundestag and related interest groups. More: <https://www.waldbrand-klima-resilienz.com/post/fr%C3%BChst%C3%BCck-und-pr%C3%A4ventives-waldbrandmanagement-in-berlin>

Date: October 4-7, 2022

AP: 4, 5

Location: Florence, Italy

Occasion: Association of Fire Ecology Conference: FIRE ECOLOGY ACROSS BOUNDARIES: CONNECTING SCIENCE AND MANAGEMENT. Comparison of international processes and adaptation to the German context.

Participants: A Held (in person), L Pronto (online)

Date: October 6, 2022

AP: 4, 5

Location: Online

Occasion: Regular meeting of the Forest Fire Expert Commission, State of Lower Saxony

People involved: L Pronto

Date: August 31 – September 1, 2022

AP: 4, 5

Location: Regensburg State Fire Service School

Occasion: Workshop "Vegetation fire - cascade effects in complex operational scenarios"

Participants: L. Pronto, FW, Bundeswehr, THW, Police, Bavarian Ministry of the Interior, BBK, OTH

Short description: 2-day workshop; distribution of WKR modules and policy brief on aerial firefighting.

Follow up: Report to the Ministry of the Interior

Date: 19 January 2022

AP: 4

Location: Online

Occasion: National Working Group: 5th Meeting UAG Tactics

Participants: L. Pronto, A. Held; UAG members (hosted by the Lower Saxony Ministry of the Interior and Sport)

Short description: Processing/preparation of draft interim report.

Follow up: 6th meeting in April

Date: 08 November 2021

AP: 4

Ort: Online /Glasgow

Anlass: COP26 Side Event: Addressing the Risk of Wildfires to Contribute to Climate Change Mitigation and Adaptation

Participants: L. Pronto; UNDESA; UNFFS etc.

Short description: Participation

Date: 30 September 2021

AP: 4

Location: Online

Occasion: Submission of contributions to the EU standardization agenda

Persons involved: L. Pronto; DG HOME EU Commission

Short description: Discussion on the need for standardization for the EU standardization agenda and the development of future funding calls (topic: forest fires)

Date: August 25-26, 2021

AP: 4

Location: State Fire Service School Regensburg (Lappersdorf)

Occasion: Workshop "Forecasting Forest Fire Challenges for the Free State of Bavaria - organized by the State Fire Service School Regensburg and the East Bavarian Technical University - Regensburg and the Bavarian State Ministry of the Interior

Participants: A. Held, L. Pronto

Short description: to proactively develop future forest fire scenarios, identify capability gaps, and find potential solutions to the challenges that Bavaria and Germany expect to face in the coming years due to forest fires. The results of the process will be directly implemented by the Bavarian Ministry of the Interior. The participating experts came from a variety of federal and state disaster management agencies, ministries, and specialist academies.

Follow up: Recommendations to the Ministry of the Interior (Bavaria)

Date: July 26, 2021

AP: 4

Location: Online

Occasion: Workshop: Forecasting forest fires - Challenges for the state of Bavaria

Participants: WKR, (Pronto and Held) + 30 participants from 6 federal states

Short description: Organized by the State Fire Service School Regensburg and the East Bavarian Technical University - Regensburg and the Bavarian State Ministry of the Interior

Follow up: Report for the Ministry of the Interior

Date: July 13, 2021

AP: 4

Location: Online

Occasion: Expert webinar "*Vegetation fire fighting Bavaria*"

Persons involved: Held, Pronto; (LFSR, LFV, DWD, OTHR, etc.)

Short description: discuss together to what extent the Bavarian fire brigades are prepared in terms of skills and equipment for pre-developed forest fire scenarios

In addition, the future cooperation between different actors in vegetation fire fighting will also be discussed

Follow up: To be followed by in-person workshop

Date: June 13, 2021

AP: 4

Location: Online

Occasion: Workshop: Forecasting forest fires - Challenges for the state of Bavaria

Participants: WKR, (Pronto and Held) + 30 participants from 6 federal states

Short description: Organized by the State Fire Service School Regensburg and the East Bavarian Technical University - Regensburg and the Bavarian State Ministry of the Interior

Follow up: Workshop Part 2 in July

WP 5 Technical recommendations for policymakers and decision-makers

Too often in Germany, we can observe how decisions regarding forest fires are made quickly, rashly, and poorly informed. The impetus for including WP5 in the WKR project was actually the political roundtables that made investment decisions during the Lübtheen forest fire in 2019. Another example of poorly informed discussions regarding forest fires in Germany is the topic of "firefighting from the air." Few decision-makers have actual experience in this area. Against this backdrop, the discussion about aircraft often seems somewhat clumsy and based on television images from the Mediterranean.

Expert recommendations for policymakers and decision-makers in the field of integrated Forest fire management is therefore important for several reasons:

1. *Risk reduction and prevention:* Experts can propose comprehensive risk assessments and preventive measures to help prevent wildfires or minimize their impact.
This includes the planning of short-term technical measures such as the creation of firebreaks, but also long-term conceptual measures such as forest tree restoration, deadwood concepts, the promotion of fire-resistant vegetation and the management of fuel materials.
2. *Coordination and planning:* Integrated wildfire management requires coordinated cooperation between different agencies, organizations, and communities. Experts can provide recommendations on how this cooperation can be optimized to ensure an efficient response to wildfires.
3. *Resource utilization:* Experts can provide advice on how to effectively utilize resources such as personnel, equipment, and finances. This helps ensure that available resources are used optimally, especially in crisis situations.
4. *Adaptation to climate change:* Climate change increases the frequency and intensity of forest fires.
Expert recommendations help to develop strategies that adapt to these changing conditions, such as adapted management practices and preparation for longer fire seasons.
5. *Protection of people and property:* Expert advice is crucial for developing measures to ensure the safety of people and the protection of property. This includes awareness-raising and education (WP6), as well as early warning, detection, and communication of warnings to the public.
6. *Nature conservation and ecosystem management:* Wildfires are part of many ecosystems, and experts can help balance necessary fire management with biodiversity conservation. This includes the use of controlled fires to promote ecological processes and as a preventative measure to reduce fire load.
7. *Policy and legal framework:* Expert recommendations support the development and implementation of laws and regulations that enable effective wildfire management. This includes advice on the formulation of building codes, land-use plans, and safety standards.
8. *Public awareness and education:* Professionals can develop public awareness programs to increase awareness of the risks and prevention measures associated with wildfires.

Overall, the WKR technical recommendations help make informed, evidence-based decisions that reduce the risk and impacts of wildfires, protect the safety and well-being of communities, and preserve natural resources and ecosystems.

The following technical recommendations were published during the project period:

- Handling aircraft, firefighting from the air
- Silvicultural measures for fire prevention
- Expert recommendation together with Forest Europe, Ministerial Conference on the Protection of Forests in Europe (EU level)
- All expert recommendations are available at: <https://www.waldbrand-klima-resilienz.com/fachempfehlung>
- Another topic within the framework of WP 5 was fire management in areas suspected of containing munitions. Although no official technical recommendation was published, a number of presentations and working documents were shared with decision-makers, which ultimately led to concrete implementation in Brandenburg.
- Two additional policy briefs were planned and partially completed and were to be published in 2024 as part of an extension of the WKR project – which unfortunately could not take place. These were a) a policy brief on the wildland-urban interface (unpublished) and b) a policy brief on fire management in inaccessible terrain (unpublished).

The technical recommendations of WP5 must be considered in conjunction with work packages 1, 2, 3, 4, 6, and 7. The activities and results from these WPs naturally also influence WP5.

WP5-relevant activities:

Date: April 25, 2023

AP: 5

Location: Berlin

Occasion: Meeting with the Brandenburg Ministry of the Interior

Persons involved: Held, LSTE, Ministry of the Interior, Doking

Short description: WKR talks in the **Brandenburg Ministry of the Interior** on the topic of fire on munitions-contaminated areas.

Date: 6-9 February 2023

AP: 5, 6, 4

Location: Barcelona, Spain

Anlass: "Communicating the importance of sustainable forest management to prevent forest fires" organized by Forest Europe, Ministerial Conference on the Protection of Forests in Europe

Persons involved: Held and Pronto, EFI, FE, CTFC, PCF

Short description: In a three-day event, FOREST EUROPE, together with the European Forest Institute, the Pau Costa Foundation, and the Center for Forest Science and Technology of Catalonia (CTFC), organized a workshop in Barcelona with foresters, researchers, journalists, and policy makers from different regions of Europe. The main objective was to discuss the growing challenges foresters face in preventing wildfires and the role of communication and the media in reporting on these events and their consequences, as well as the practices being implemented to create more resilient forests. Held and Pronto presented the WKR case studies as an example and reported on their recent collaboration with the PhD students of the Pyrolife project to produce the FE/WKR Policy Brief "Reducing wildfire risk through Sustainable Forest Management."

Date: January 24-26, 2023

AP: 5, 1, 4

Location: LSTE Brandenburg/Online

Occasion: Working meeting of the UAG Tactics and Training of the national forest fire working group

Persons involved: L Pronto, UAG

Date: December 12, 2022

AP: 5, 6

Location: Online

Occasion: Discussion on forest fires with AXA forest fire insurance

Persons involved: AXA, A Held

Date: December 7, 2022

AP: 5

Location: Online

Occasion: Policy Brief workshop with PyroLife PhD Program

Participants: L. Pronto, A Held

Short description: Further development of WKR/FE technical recommendation

Date: November 21, 2022

AP: 5, 4

Location: Online

Occasion: Planning meeting with WKR, BMEL and Forest Europe

Participants: L. Pronto, A. Held, **Short description:**

Planning of forest fire communication workshop and expert recommendation

Date: November 8-10, 2022

AP: 5, 4

Ort: LSF Regensburg

Occasion: Working meeting of the UAG Tactics and Training of the national forest fire working group

People involved: L. Pronto

Date: 24-28 October 2022

AP: 5, 4

Location: Antalya, Turkey (General Directorate of Forestry of Turkey)

Anlass: Workshop von Silva Mediterranea (FAO) and Forest Europe (Ministerial Conference on the Protection of Forests in Europe):

"Develop, adopt and transfer innovative solutions and actions to prevent and control wildfires".

People involved: L. Pronto, A Held

Short description: WKR supported this high-level initiative by co-organizing and selecting experts together with Forest Europe and the BMEL – as the German government's contribution to the RD process. Alex delivered a presentation on "Innovation in Fire and Forest Management: From Tactics, Technology, and Tools to the Human Factor," and Lindon delivered a presentation on "Forestry and Disaster Management – A Shared Responsibility."

Follow up: Forest Europe /WKR policy brief

Date: 5.May 2022.

AP: 5, 6

Location: Online

Occasion: Risk Communication Workshop

Participants: FVA, WKR, PyroLife Project

Date: February 23, 2022

AP: 5, 6, 7

Location: Online

Occasion: Rosenbaur Wildfire X-Change Seminar

Participants: L. Pronto, A. Held

Brief description: Development and impacts of wildfires; vegetation stress and climate change; prevention at all levels (government, fire service, forestry, community; including building standards, land management, risk assessment, fire simulation); early detection (through sensors, satellites, drones, humans, etc.); tactical wildfire suppression (fire behavior, risk assessment, backfires, rapid deployment of resources, navigation in the forest, aerial wildfire suppression, digitalization in wildfire suppression); extinguishing methods (technical developments, dry and wet extinguishing); training (challenges, simulation, virtual)

Follow up: Founding of International Experts Network

Date: 19-20 January 2022

AP: 5, 3

Location: Online

Anlass: (High-Level) Expert Group meeting on the Forest Risk Knowledge Mechanism

Participants: L. Pronto (notetaker/presenter), A. Held; EFI; Forest Europe; various international organizations/governments

Short description: Discussion of gaps, examples of best practices, and areas for investment to address forest risks across Europe. Forest fire prevention and management are frequently highlighted.

Follow up: Publication of workshop outcomes

Date: 18 January 2022

AP: 5, 3

Location: Online

Anlass: Joint Fire-IN /MEDEA event

Participants: L. Pronto; Fire-IN and MEDEA project consortiums, solutions providers, skateholders

Short description: Identification of future common capability gaps in the European fire and rescue scene, with presentations from solution providers and discussions with practitioners/other stakeholders.

Date: 14 January 2022

AP: 5, 4

Location: Online

Event: Forest Europe + EFI

Participants: L. Pronto; J. Haertel (Forest Europe Secretariat)

Short description: Discussion on the design of forest fire working groups for Forest Europe and the pilot phase to be supported by WKR

WP 6 Educational work and awareness raising

Website

Within the first few months of the project, a website was created, which was regularly updated on a daily or weekly basis throughout the project phase. The website continues to be maintained and updated. The website was continuously updated—not only with project content, but also with relevant publications, news from the network, and a listing of media collaborations by the WKR wildfire experts.

The amount of content and reach of the project website is unique in the German-speaking world. Combined with the project's social media activities and public relations work,

With press, radio, and television coverage, the website has established itself as a central element in awareness-raising efforts in Germany.

www.waldbrand-klima-resilienz.com

The channels and reach of the European Forest Institute, as well as the WKR network, are also used to raise awareness and promote project content.

Raising awareness about WKR activities also had an international impact, which in part secured more participants and expanded the network for both project activities and multipliers

Here are some excerpts from Wix and Google Console website traffic analytics.

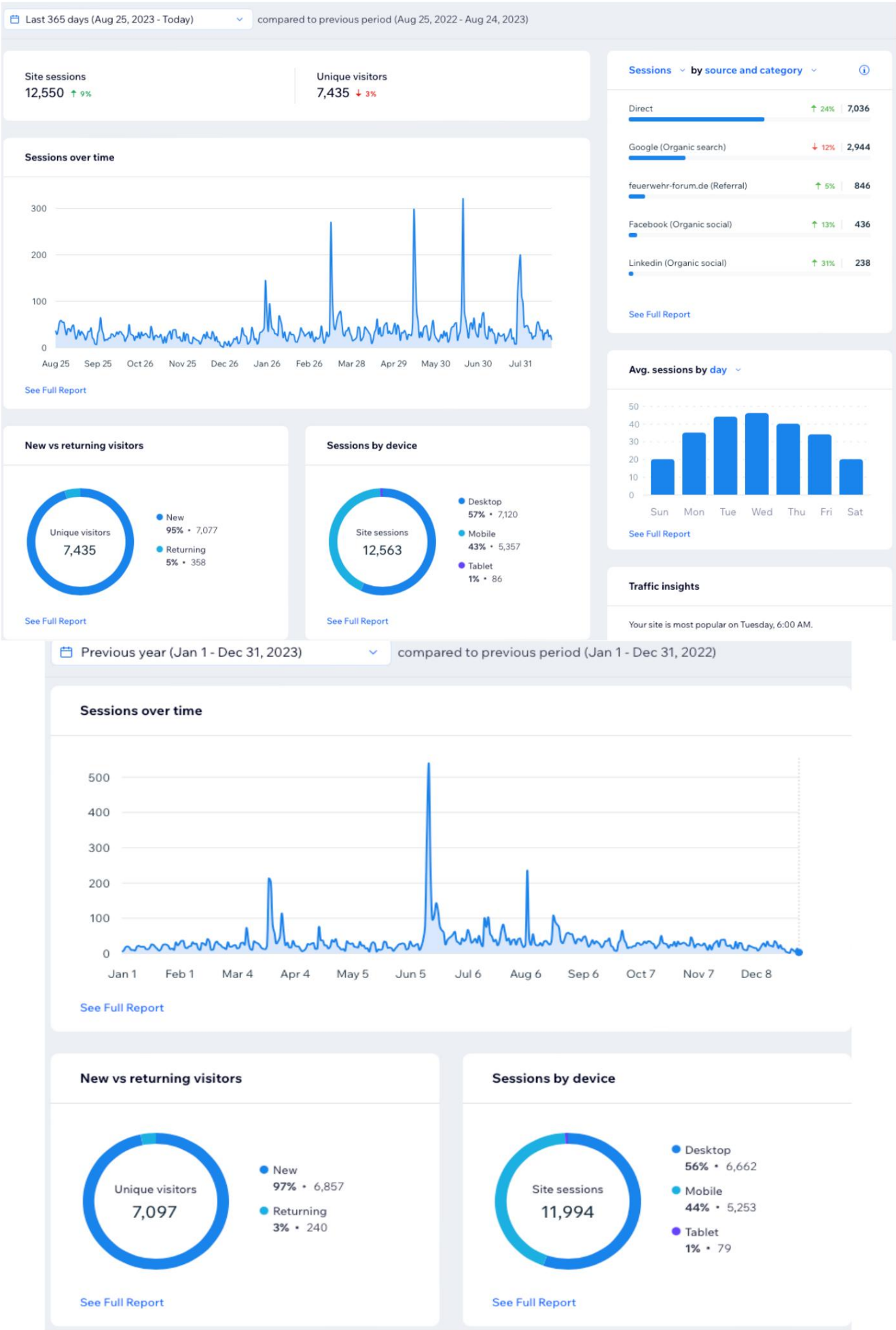
Previous year (Jan 1 - Dec 31, 2023)

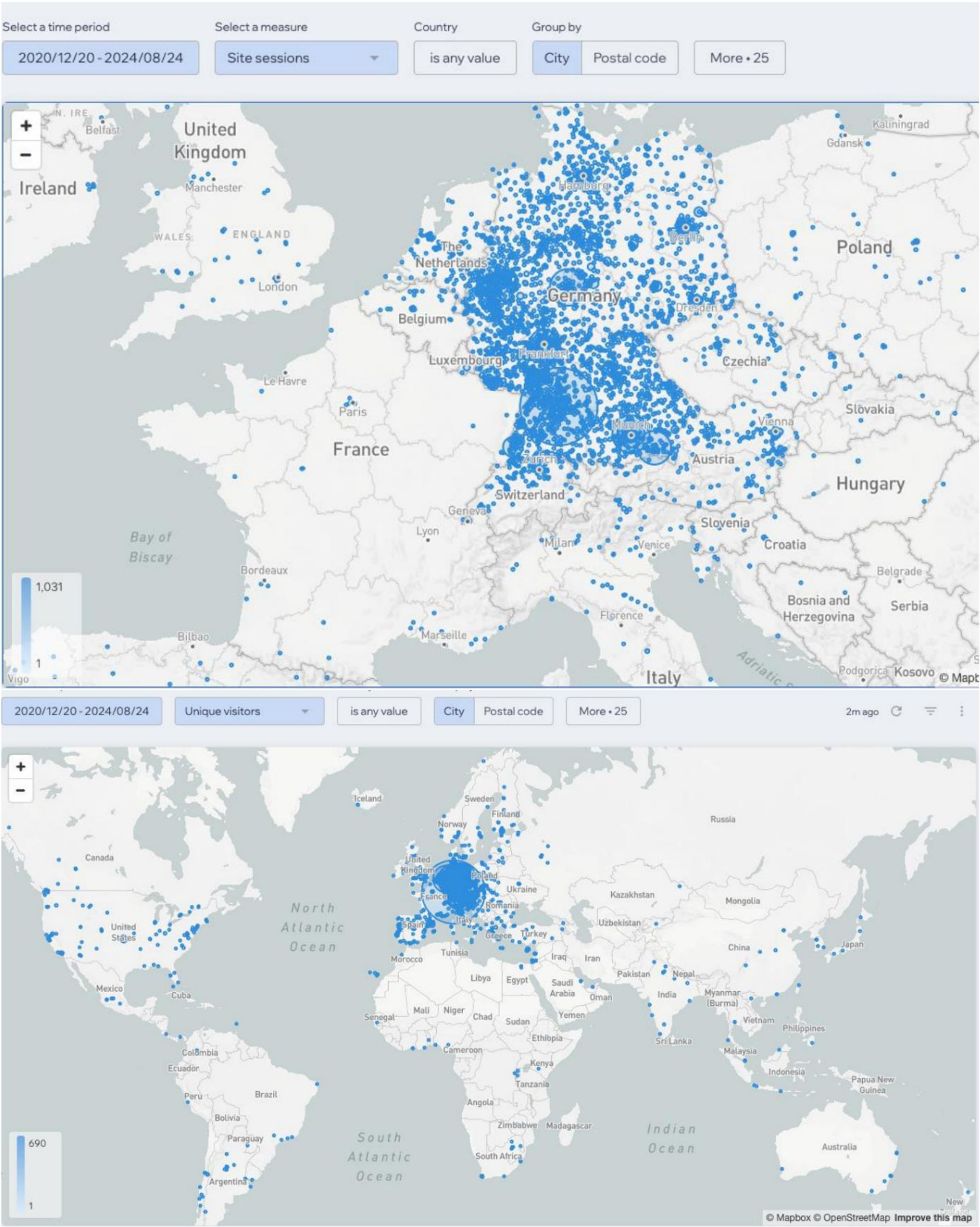
Group by day

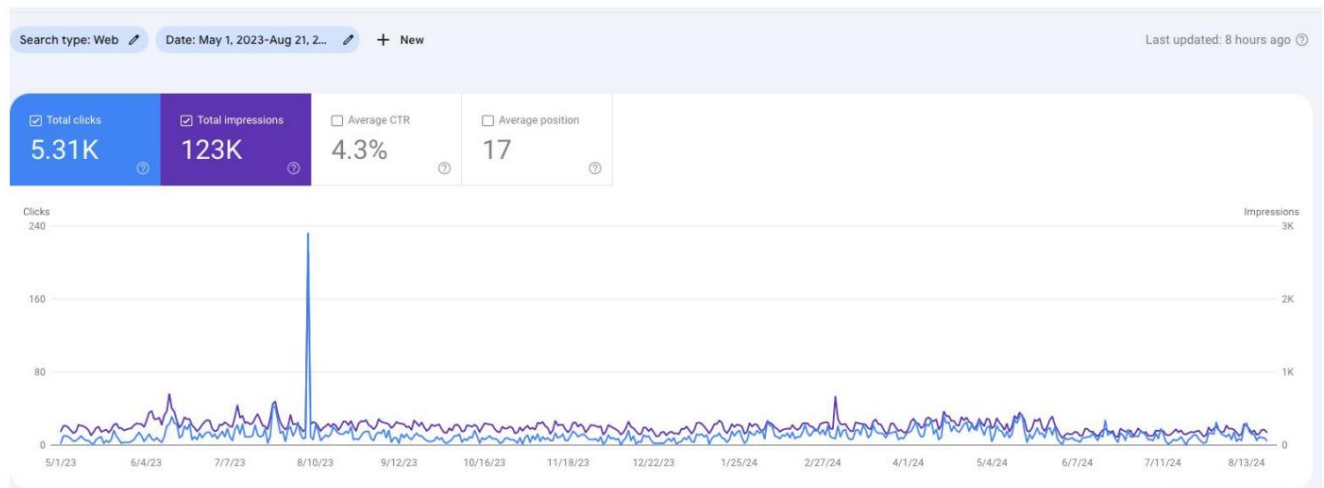
Filter

Customize

Date	City	Country	Site sessions ⓘ	Unique visitors ↓	Page views
Summary			11,985	7,097	25,119
Jun 15, 2023		DE	50	41	59
Jun 15, 2023	Berlin	DE	26	23	42
Jun 14, 2023		DE	23	22	27
Jun 15, 2023	Frankfurt am Main	DE	23	19	60
Nov 19, 2023	Leinefelde-Worbis	DE	18	18	18
Jun 15, 2023	Stuttgart	DE	21	18	45
Aug 7, 2023		DE	17	17	18
Oct 14, 2023	Uder	DE	16	16	16
Jun 15, 2023	Vienna	AT	16	15	25
Jun 14, 2023	Frankfurt am Main	DE	16	15	34
Mar 21, 2023	Berlin	DE	16	15	27
Sep 27, 2023	Prague	CZ	15	14	25
Apr 17, 2023		DE	16	14	32
Nov 17, 2023	Leinefelde-Worbis	DE	14	14	14
Mar 22, 2023	Berlin	DE	14	13	45
Nov 20, 2023	Leinefelde-Worbis	DE	13	13	13
Jun 14, 2023	Berlin	DE	16	13	26
Mar 21, 2023		DE	13	13	23
Mar 22, 2023		DE	14	12	23
Jun 14, 2023	Munich	DE	13	12	38
Jul 17, 2023	Frankfurt am Main	DE	11	11	18
Jun 15, 2023	Bestwig	DE	11	11	22
Jun 19, 2023		DE	10	10	12
Mar 28, 2023		DE	12	10	20
Apr 24, 2023	Stuttgart	DE	12	10	29
Jun 15, 2023	Cologne	DE	12	10	33
Apr 17, 2023	Frankfurt am Main	DE	11	10	51
Oct 20, 2023	Heilbad Heiligenstadt	DE	10	10	10
Jun 15, 2023	Munich	DE	11	10	43
Oct 29, 2023	Heilbad Heiligenstadt	DE	9	9	9
Jun 16, 2023		DE	11	9	17







Media work

Summary of interviews by year:

2023: Interviews: Print/Online (31), Radio (11), TV (19), Podcasts (2), Articles (1), Videos (3)

2022: Interviews: Print/Online (41), Radio (14), TV (23), Podcasts (3), Articles (1)

2021: Interviews: Print/Online (17), Radio (12), TV (3)

2020: Interviews: Print/Online (7), Radio (3), TV (2)

Interviews in reverse order:

- Interview: December 1 (2023), **Wendland-net**: Training: "So that the forest does not go up in smoke" (Held)
- Interview: September 14 (2023), **BILD**: "Why are fires different today than they used to be?" (Pronto)
- Interview: September 11 (2023), **Quarks**: Forest fires caused by climate change – or not? (Hero)
- Podcast interview: September 9 (2023), **ARD-Weltspiegel**: Mega fire in paradise - Hawaii after the Brand (Held)
- Podcast-Interview: 31. August (2023), **EURACTIV** Beyond the Byline Podcast: „What can we learn from this summer of wildfires?“ (Pronto)
- TV interview: August 28 (2023), **ARD Tagesschau**: Why is Canada burning so fiercely? (Held)
- Interview: 26 August (2023), **Neue Zürcher Zeitung**: Europe in flames: More cooperation on Supporting overwhelmed firefighters (hero)
- Interview: August 26 (2023), **BILD am Sonntag** (online and print): "From Africa to Australia: Why are fires burning more and more often around the world?" (Pronto)
- Interview: August 22 (2023), **Editorial Network Germany**: Fire ecologist Alexander Held: Why do islands burn so often? (Hero)
- TV Interview: Breaking News Segment: August 22 (2023), **WELT – Panorama**: "Forest fires in Greece – 18 burned bodies discovered in Dadia National Park" (Pronto)
- Interview: August 18 (2023), **Süddeutsche Zeitung**: "In a healthy forest, it is not so easy to light something with a cigarette" (Held)
- Interview: August 11 (2023), **FORUM Magazine** (Issue 33, pp. 32-35): "Global Warming – Heat" – Titelseite (Pronto)
- Radio: August 2 (2023), **Bremen Zwei**: After the fire: How long does nature need to recover? (Held)
- TV interview: August 2 (2023), **Euro News** (published in various countries and languages): "Wildfire prevention: How Europe plans to handle blazes in the years ahead" (Pronto)

- Interview: July 27 (2023), **Perspective Daily**: Wildfires: Why we must learn to reconnect with the Fire to live (Hero and Pronto)
- TV interview: July 26 (2023), **Bayern2**: Fire expert: People are the ignition source (Hero)
- TV interview: July 25 (2023), **Phoenix**: Alexander Held on the devastating fires in Mediterranean (Hero)
- TV interview: July 25 (2023), **ZDF Moma**: Expert: "Extreme fires" are increasing (Held)
- TV interview: July 25 (2023), **WDR aktuell**: Interview with EFI fire expert (6:48 min) (Held)
- Radio interview: July 25 (2023), **Hessischer Rundfunk**: Hourly news segment "HR-Info" (Ready)
- TV interview: July 24 (2023), **TRT World News**: Forest fires in Greece out of control (Held)
- Interview: July 24 (2023), **T-Online**: "This is a life-threatening trap" (Held)
- TV interview: Evening news segment: July 23 (2023), **Channel 4 News** (UK): about Mediterranean fires (Pronto)
- TV Interview: Breaking News Segment: 23 July (2023), **BBC World News** (UK): about the fires on Rhodos (Soon)
- Interview: July 23 (2023), **Süddeutsche Zeitung**: "Fighting the flames with a pickaxe and helicopter" (Ready)
- TV-Interview: 23. Juli (2023), **WION** (India): „Europe wildfires destroy homes and force evacuation" (Ready)
- Radio interview: July 23 (2023), **BBC Radio London** (UK): Morning news segment on Mediterranean fires (Pronto)
- Radio interview: July 23 (2023), **BBC Radio Scotland** (UK): Morning news segment on Mediterranean fires (Pronto)
- TV interview: Evening news segment: July 21 (2023), **Bayerischer Rundfunk**: (Abendschau-South) (Pronto)
- TV interview: July 20 (2023), **DW News**: Hourly global news broadcast (Pronto)
- Interview: July 14 (2023), **Focus Online Earth**: "Germany finally needs a strategy to combat the drought fires" (Pronto)
- Interview: 13. Juli (2023), **Financial Times** (UK): „Extreme wildfires are here to stay. Can human beings really fight them?" (Pronto)
- Interview: June 22 (2023), **Krautreporter**: The dispute over forest fires and climate, explained in an understandable way (Pronto and Held)
- Radio interview: June 13 (2023), **Hessischer Rundfunk**: "HR-Info" (Pronto)
- TV interview: June 9 (2023), **Servus TV Germany**: "Good Evening Germany" (Pronto)
- Interview: June 8 (2023), **Der SPIEGEL**: Inadequate forest fire protection (Hero)
- Interview: June 7 (2023), **AFZ-der Wald**: Against the Fire (Hero)
- TV-Interview: 27. May (2023), **Polsat News** (PL): "Lubuskie. International Training of the Security Services Rescue" (Pronto)
- Interview: April 26 (2023), **Der Spiegel**: "Is Croatian mine clearance technology suitable for Brandenburg's forests?" (Held and Pronto)
- Interview: April 22 (2023), **Nuremberg News**: Forest fire: Forestry, fire service and mountain rescue are in demand (hero)
- Radio interview: March 21 (2023), **SWR**: How Karlsruhe prepared for forest fires (Held)

- TV interview: March 21 (2023), **BR television**: Four out of five trees are sick (Held)
- TV interview: March 21 (2023), **ARD Tagesschau**: How to prevent forest fires (Held)
- Interview: March 21 (2023), **Globe Echo**: New Concepts: How to Prevent Forest Fires (Hero)
- Interview: February 15 (2023), **Oberoerdingen Volunteer Fire Department**: Participation in the colloquium of the Baden-Württemberg Forest Research Institute on the topic of "Forest Fires" (Held)
- Interview/Video: February 1 (2023), **FNR**: Forest Fire Prevention (Hero)
- Interview/Video: February 1 (2023), **FNR**: Is deadwood fuel? (Held)
- Interview/Video: February 1 (2023), **FNR**: Contemporary Forest Fire Management (Held)
- Interview: January 4 (2023), **AFZ The Forest**: Climate change requires vision in fire management (Held)
- Radio interview: January 3 (2023), **Deutschlandfunk**: Growing danger of forest fires in Europe (Held)
- Radio interview: "Growing forest fire risk in Europe," **Deutschlandfunk**, December 21, 2022. (Ready)
- Interview: October 28, 2022, **Holzblatt**: "Forest fires: Making an emergency call is not enough" (Held)
- Interview: October 11, 2022, **FNR**: "Managing forest fire risks all year round" (Held)
- TV interview: 26 September 2022, **Sat.1** regional: "Heide in Flames: Large fire brigade exercise in Lüchow" (hero)
- TV interview: September 24, 2022, **Elbe-Jeetzel-Zeitung**: "At Trebel: Fighting fire with fire" (Hero and Pronto)
- Update: 23. September 2022, **Nowa Gazeta LejŃa** (PL): "Record-breaking fires in Europe" (Pronto)
- Radio interview: September 20, 2022, **Deutschlandfunk Kultur**: "A journey through the forest fire summer" (Held)
- Interview: 16 September 2022, **Lübecker Nachrichten**: "How can we protect our forests in the future better protect against fires?" (Hero)
- Mention: September 9, 2022, **Der Spiegel**, No. 37 (print edition): "Winged Fire Brigade" (Pronto)
- Interview: 7 September 2022, **Der Spiegel+**: "Canadair 415 firefighting aircraft: Does Germany need this machine?" (Pronto)
- Interview: 7 September 2022, **Der Spiegel+**: "In the Harz Mountains, a smoldering fire is eating its way through the ground – How can it be deleted?" (Pronto)
- Interview: August 28, 2022, **OVB - Oberbayerisches Volksblatt** (online and print): "Amidst blazing flames: How a firefighter from Aschau im Chiemgau fights forest fires" (Pronto)
- TV interview: August 29, 2022, **ZDF-Hart aber Fair** (studio guest): "The drought of the century: Are we currently experiencing our future?" (Held)
- Interview: August 21, 2022, **Der Spiegel+**: "How to best fight forest fires with money" (Ready)
- Interview: 16 August 2022, **Mitteldeutscher Rundfunk** (MDR): "What can we learn from other Forest fire regions in the world?" (Pronto)
- Interview: 12 August 2022, **Riff Reporter**: "After the forest fires: How will the resilient forest of the future look like?" (Hero)
- Interview: August 10, 2022, **Badische Neueste Nachrichten**: "Agriculture Minister Peter Hauk learns about forest fire fighting in Stutensee" (Held)
- Podcast interview: 10 August 2022, **STERN Podcast**: "Germany is now a forest fire country – it knows

only hardly anyone" (Hero)

- Podcast interview: August 10, 2022, **WDR-5**: "Forest fire protection concept for NRW 'makes sense'" (Held)
- Interview: 10. August 2022, **ATL** (SE): "Strong increase in forest fires in Europe" (Pronto)
- Interview: 10. August 2022, **ATL** (SE): „Forest fires are increasing: 'What we warned about is now reality'" (Ready)
- TV interview: 4 August 2022, **Tag24**: "Nature conservation versus fire protection: dispute over dead wood in a National Park (Hold)
- Interview: 3 August 2022, **NZZ**: "Germany is suffering from increasingly severe forest fires – a unsolvable problem in times of climate change?" (Held)
- Radio interview: August 3, 2022, **Deutschlandfunk**: "Only ecological forest conversion can really help" (Held)
- Interview: August 2, 2022, **Der Spiegel**: "Calling emergency services and hoping for rain is not a strategy" (Held)
- TV interview: July 28, 2022, **MDR**: "Smoldering embers are a big problem" (Held)
- TV interview: July 27, 2022, **Phoenix**: "Alex Held is a guest on 'Der Tag' and explains why Germany is still lagging behind in the fight against forest fires" (Held)
- Interview: July 27, 2022, **Berliner Zeitung**: "Forestry expert: New strategies for forest fire fighting needed" (Held)
- TV interview: July 27, 2022, **Tag24**: "Forest fires in Saxon Switzerland and the Czech Republic: Situation tense, but not out of control" (Hero)
- Interview: July 27, 2022, **Augsburger Allgemeine**: "Forestry expert: New strategies for forest fire fighting needed" (Held)
- TV interview: July 27, 2022, **ZDF**: "Fires in Europe – The Fight Against Fire" (Held)
- TV interview: July 26, 2022, **WDR aktuell**: "How can forest fires be prevented? WDR Aktuell interviewed Christian Schmidt and Maria Schloßmacher in the Kottenforst" (Held)
- Interview: July 26, 2022, **Oberhessische Presse**: "Fire ecologist calls for forest fire commissioner at federal level" (Held)
- Radio interview: July 26, 2022, **Das Inselradio Mallorca**: "Forest fire risk on Mallorca extremely high" (locksmith)
- Radio interview: July 26, 2022, **Deutschlandfunk**: "Researchers: Learning to live with fire" (Held)
- Interview: 26 July 2022, **RND**: "Fire ecologist Alexander Held: How do we learn to cope with forest fires live?" (Hero)
- TV interview: 26 July 2022, **ZDF-Drehscheibe**: "The risk of forest fires is increasing – in the model region In Baden-Württemberg there are concepts to protect the forest" (Held)
- TV interview: July 24, 2022, **rbb 24**: "How can a fire department save water during forest fire operations? Visit to the special 'Vegetation Fire' unit and WKR project partners" (Semsch)
- TV interview: 22 July 2022, **WDR 5**: "More political will to coordinate the Forest fire fighting between firefighters and foresters – demands Alexander Held" (Held)
- Interview: July 22, 2022, **Süddeutsche Zeitung**: "Europe in Flames: Alexander Held in an interview about an extraordinary year" (Held)
- TV interview: July 20, 2022, **NDR**: "Heat records: many fires in the north" (Held)
- Interview: July 19, 2022, **Stern**: "Highest forest fire warning level in Germany. One spark is enough. Are Are we prepared? Not at all, says one expert" (Held)

- TV-Interview: 21. Juli 2022, **BBC World News**: „Europe's heat emergency“ (Pronto)
- Interview: 10 July 2022, **MDR Sachsen-Anhalt**: “We must make an effort not to overburden the fire brigades to leave oneself to oneself” (Held)
- Radio interview: June 27, 2022, **Deutschlandfunk**: “Forest firefighting: counterfire and other innovations – Interview with A. Held” (Held)
- Article: Summer issue 2022, **German Forest Protection Association**: “Forest conversion as a solution” (Held)
- Interview: Summer issue 2022, **Beautiful Life**: “They will go through fire for your forests” (Held)
- Radio interview: 24 June 2022, **BR24**: “No, a piece of glass probably cannot start a forest fire cause” (Hero)
- Interview: June 21, 2022, **Bild online**: “Germany is on the way to becoming a forest fire country” (Held)
- Interview: June 20, 2022, **Forest Climate Fund**: “European teamwork to combat forest fires” (Held)
- TV interview: June 20, 2022, **MDR Aktuell**: “We can still work on strategy and tactics” (Held)
- Interview: June 20, 2022, **Der Spiegel**: “Germany's powder keg” (Hero)
- Interview: 20 June 2022, **Redaktionsnetzwerk Deutschland (RND)**: “Germany is a bit Arrogant about forest fires.” Online and in numerous print editions. (Pronto)
- TV interview: January 1, 2022, **DW News**: “Forest fires like those in Colorado also in winter” (Pronto)
- Interview: 30 May 2022, **Focus online**: “Fire ecologist warns of forest fire summer - these are the Hotspots in Germany” (Held)
- Interview: 29. Oktober 2021, **Vallfirest** (en): „Wildfire Diaries Episode VII“ (Held)
- Radio: 28 October 2021, **Hessischer Rundfunk**: “Forest fire danger – how our forests are changing must become” (Hero)
- Interview: 26. September 2021, **CBC** (en): „After devastating European fire season, experts call for new approach to protecting forests“ (Held)
- Interview: 4. September 2021, **The INDEPENDENT** (en): „What can we learn from Europe's blazing summer?“ (Held)
- Interview: 30. August 2021, **Bloomberg News** (en): „A Tale of Two Wildfires: Lessons From Europe's Blazing Summer“ (Held)
- Interview: 27. August 2021, **Brussels Morning Newspaper** (Magazine cover story) (en): „Wildfires are the most human of natural disasters“ (Pronto)
- Interview: August 22, 2021, **Watson**: “Negligence, ignorance, greed: humans cause most wildfires” (Pronto)
- Interview: August 17, 2021, **FAZ**: “These machines alone will not extinguish the fire” (Held)
- Interview: 15 August 2021, **Watson**: “Fire expert warns: ‘Forest fires are becoming more and more common in Germany more to the challenge’” (Pronto)
- Radio: 13 August 2021, **Deutschlandfunk Nova**: “How forests can be better protected from fires can” (Pronto)
- Radio: August 12, 2021, **Deutschlandfunk Nova**: “How do firefighting aircraft work?” (Pronto)
- Radio: August 12, 2021, **WDR5**: “Making forests fireproof” (Held)
- Radio: August 11, 2021, **Deutschlandfunk Wissenschaft**: “Satellite data makes firefighting easier” (Held)

- Interview: August 11, 2021, **t-online**: "The same mistake is always made after forest fires." (Held)
- Interview: August 11, 2021, **Süddeutsche Zeitung**: "Fight fire with fire?" (Held)
- Interview: August 11, 2021, **Salzburger Nachrichten**: "How emergency services fight forest fires" (Ready)
- Interview: August 11, 2021, **Aktiv-online**: "Ororatech detects forest fires with satellites" (Held)
- Interview: 10 August 2021, **Hamburger-Abendblatt**: "Frightening images – Why is half of the Planet?" (Held)
- Podcast: August 9, 2021, **WDR5**: "Forest fires – does this affect us all?" (Held)
- Radio: August 9, 2021, **Hessischer Rundfunk**: "HR-Info – Forest fires in Greece" (Pronto)
- Radio: August 9, 2021, **SWR2**: "Forest fire expert: 'This is not a normal forest fire summer'" (Held)
- Radio: August 6, 2021, **SWR Aktuell**: "Forest fire expert on fires in Southeast Europe: 'The smoke is the worst'" (Held)
- Interview: 5. August 2021, **THE ECONOMIST** (en): „How are wildfires fought?“ (Held)
- TV interview: August 3, 2021, **DW-TV**: "Extreme heat and fires in the Mediterranean" (Pronto)
- Radio: August 2, 2021, **Deutschlandfunk Kultur**: "This could become the new normal" (Held)
- Interview: August 1, 2021, **Die ZEIT**: "Living in the countryside means living with fire" (Pronto)
- Radio: 14 July 2021, **Deutschlandfunk Nova**: "Protection against forest fires – How goats fight forest fires prevent" (hero)
- Interview: June 29, 2021, **EJZ**: "Gartow Forest: Depriving the forest fire of fuel" (Held)
- TV interview: June 24, 2021, **NDR**: "Fuel buffer zones to prevent forest fires" (Held)
- Interview: May 31, 2021, **MAZ**: "Forest fires in Brandenburg in full swing" (Held)
- Interview: May 4, 2021, **STATE FORESTRY**: "Cooperation in forest fire prevention" (Held)
- Radio: 3 May 2021, **ANTENNE**: "Forestry office gets fire tanks for pick-up trucks: Forests to dry" (hero)
- Interview: 29. September 2020, **Bloomberg News** (en): „Europe's New Trick Against Wildfires: Let It Burn" (Held)
- TV interview: September 28, 2020, **SWR**: "Forest fire prevention: model project in Baden-Württemberg" (Held)
- Interview: 25 September 2020, **Süddeutsche Zeitung**: "Baden-Württemberg prepares for forest fires" (Held)
- Interview: September 17, 2020, **BILD**: "Forest fires in Brazil out of control!" (Pronto)
- Radio interview: 16 September 2020, **Deutschlandfunk Nova**: "Fire weather, monocultures and Bark beetles promote the spread of forest fires" (Held)
- Interview: 16 September 2020, **Frankfurter Rundschau**: "Climate change as an accelerant: Wildfires in the US could last for months" (Pronto)
- Interview: 15 September 2020, **Süddeutsche Zeitung**: "Plenty of fuel" (Held)
- TV-Interview: 14. September 2020, **France24 The Debate** (en): „Why is the West Coast burning?“ (Held)
- Interview: September 12, 2020, **BILD**: "Fire expert: 'Actually, there should be more fires!'" (Held)
- Radio interview: 27 August 2020, **SWR2**: "How forest fires can be kept small or avoided

can" (hero)

- TV interview: August 22, 2020, **Bild TV**: "Fiery hell in California: 5 dead" (Pronto)
- Interview: August 21, 2020, **BILD**: "Corona makes the fires particularly dangerous this year" (Ready)

WKR-Blog Posts (41)

March 21, 2024: [Forest fire prevention and operational preparation create synergies and added value](#)

In the Hartheim am Rhein municipal forest (gravel, sand, pine, dry!!), measures for forest fire prevention and fire department deployment preparation are being implemented in close coordination with the forest district manager (TS) and the senior forester (HA). HA has participated in several WKR training courses on the topic and has been able to gather input from various federal states. TS is writing his trainee thesis on the implementation of these measures, which he accompanies and supports with the WKR initiative.

March 14, 2024: [Burn, Eat, Sleep, Repeat: You learn how to handle fire in Handling Fire 2.0!](#)

As in previous years, we were able to use planned controlled fires in open land nature conservation in cooperation with various actors not only to practice the use of fire in nature conservation, prevention and as a tactical tool, but also to learn in practice the influencing factors on fire behavior.

In addition, we were able to demonstrate operational command and coordination, communication on and with different radio systems, the use of flame cans, fire beaters, backpack sprayers and various models of 4x4 pick-up forest fire fighting vehicles with professional and self-built skid units. practice.

January 25, 2024: [WKR project period comes to an end - WKR initiative lives on!](#)

The WKR project officially ends in January 2024. Over the past three and a half years, we have been able to collaborate with, inspire, coordinate, supply, and learn from an incredible number of amazing people and organizations. The project's impact extends far beyond the actual products and results. Many relationships, networking activities, unplanned positive developments, and projects will remain "in the dark," as many contributions remain unpublished or WKR's contribution to building wildfire capacity in some German states is not immediately visible. WKR's influence has also transcended borders, languages, and cultures.

January 23, 2024: ["The value of our knowledge multiplies when we share it with others" EoE in South Africa.](#)

We were able to arrange a stay in South Africa for two participants from Germany. Nico Semsch, unit leader of the *Special Forces Vegetation Fire* (SKV Zehdenick), member of the Waldbrandteam eV (Forest Fire Team), and guest lecturer/multiplier at the LSTE Brandenburg is completing several months of training and advanced training as part of a helipad crew in and around Cape Town. Jakob Held, "the next generation of firefighters," also gained experience for a few weeks as part of a helipad crew. We particularly support young professionals, as we fear a shortage of skilled workers in forest fire management as well!

November 29, 2023: [So that your forest assets don't go up in smoke!](#)

With the title "So that your forest assets do not go up in smoke!", the Celler Land, Ostheide and Südheide forestry cooperatives, represented by Managing Director Volker Schulte, invited two representatives from the boards of each of the 19 local forestry cooperatives, the forestry staff of the Südostheide Chamber of Agriculture Forestry Office, as well as the district administrators and district fire chiefs of the districts of Celle, Gifhorn, Lüchow-Dannenberg and Hanover, on November 23, 2023.

22. September 2023: [Czech Wildfire Academy](#)

Between September 15th and 17th around 100 firefighters and senior officers gathered in the mountains of Olešnice on the Czech-Poland border for an intensive training camp organized by PrioZs Fire and Rescue with training support from the European Forest Institute and the Pau Costa Foundation, and further support from further private companies. Most attendees were Czech volunteer firefighters, a group of professional firefighters and officers, and a contingent of high-ranking officers/colonels from Slovakia. The training was inspired by and organized in a format based on SNEPs Forest Camp (Poland, 2021 and 2022) co-organized by the WKR project.

August 14, 2023: [New skid units for Potsdam](#)

Thanks to trustful networking, the WKR project was able to provide the Potsdam Mittelmark district with two so-called *skid units* from the manufacturer Vallfirest, developed for forest fire fighting on 4x4 pick-up vehicles.

14. Juni 2023: [Fuelling the Policy – From groundwork to an international audience](#)

More than 100 participants from 22 countries gathered in the Polish forest to attend the SNEP (EN: Association of Independent Fire Expert) Forest camp, an initiative supported by EFI, PCF and IAWF. This is not only a great number, but it also proves the incredible teamwork of the forest fire community, where instructors share 46

knowledge and capacity to train future experts and trainers of forest fire management. The forest fire community consists of more than people actually fighting the fire. Therefore, SNEP organized the forest camp in Gubin, close to the German border and invited next to firefighters, foresters, policy advisors, scientists, students, and journalists. The Exchange of Expert took place between 27-29 May, and was co-organized by the WKR project the second year in a row.

April 26, 2023: Experts call for better forest fire fighting

As the first signs of spring reach Berlin and vegetation awaits sunshine, the Bundestag's Committee on Internal Affairs is holding a public hearing on forest fire protection. Experts invited by the parliamentary groups called for improvements in forest firefighting. The consensus among the experts is one, but not the only, positive development, says Alexander Held.

The WKR project manager and forest fire expert at the European Forest Institute was invited by the Alliance 90/The Greens party and emphasized in his statement that better training and thus additional investments in civil protection are necessary.

March 10, 2023: You learn how to handle fire by handling fire!

On the subject of [fire behavior](#) and other topics, we have therefore [developed training modules](#) in the WKR project [_____](#). It also explains when using a leaf blower is effective. And, interestingly, that water isn't always needed to fight fires.

February 8, 2023: Breakfast & Forest Fire Management in Berlin

The German Forest Protection Association (SDW) invited WKR project manager Alex Held to give a lecture on forest belt management as part of its "Forest Knowledge" event series on February 3 in Berlin. This was the second event in this series and is aimed at members, speakers, and staff of the German Bundestag and related interest groups.

31. Januar 2023: "Future challenges of forest fires in Germany" Workshop results of June 17, 2022.

On June 17th 2022, An expert workshop was conducted on the future challenges relating to wildfires in Germany (coming 10-15 years) as a side event of the INTERFORST Trade Fair in Munich Germany. EFI (WKR) project with FVA and the [Ostbayerische Technische Hochschule Regensburg](#) (OTH Regensburg) in cooperation with the [Pau Costa Foundation](#), coordinator of the FIRE-IN Thematic Working Group "[Landscape Fire Crisis Mitigation](#)" developed a workshop to estimate the likely major challenges for landscape fire management in Germany in the next 10-15 years.

November 14, 2022: WKR impact in practice

On 14 November, the WKR project welcomed a delegation of district managers from the [State Forestry Authority of Saxony-Anhalt](#) to [discuss the topic](#) of "Forest Fire Management in State-Owned Enterprises" with WKR Project Manager Alexander Held. The colleagues traveled to Baden-Württemberg on Sunday.

26. September 2022: Exchange of Experts in Croatia

Between September 11-16 the WKR project facilitated an exchange of experts /study tour to Croatia. The week began in Zagreb with the world-leading demining company Dok-Ing to explore the application of humanitarian remote-controlled demining vehicles for fire management on terrain contaminated by unexploded ordinance (UXO). From Zagreb the group travelled to Croatia's national interagency wildfire command center in Divulje, the Croatian aerial firefighting squadron at the Zadar airbase, and Rijeka with the County Firefighters Association and Rijeka Fire Brigade.

August 4, 2022: 36° and it's getting hotter: Inside Interforst, outside forest fire

WKR exhibited at the BMEL booth at Interforst in Munich. From August 17-20, everything at Messe München was focused on forests. Interforst is the leading trade fair for forestry and forestry technology and brings together over 300 exhibitors and 31,000 visitors from almost 60 countries. At the invitation of the Federal Ministry of Food and Agriculture (BMEL), three projects were selected to present their projects at the ministry's booth at Interforst. The Forest Fire Climate Resilience Project, funded by the Forest Climate Fund and based at EFI Bonn, was one of these projects.

25. Mai 2022: Exchange of Experts in Polen- Teamwork at its best!!

When over 100 motivated forest fire experts of today and tomorrow from seven nations come together in Poland to practice various forest firefighting tactics at full speed for three days at an Exchange of Experts (EoE), and the weather plays along despite the forecast, the weekend has gone perfectly!

18. Mai 2022: NEMAUSUS: Spain – Portugal Study Tour and Exchange of Experts (EoE)

In the frame of a cooperation between the [Advanced Fire Analyst Network](#) project and the NEMAUSUS project of the Union Civil Protection Mechanism, a weeklong study tour in Spain and Portugal was organized by the [Pau Costa Foundation](#) and the [Waldbrand Klima Resilienz \(WKR\) project](#). Spanish and Portuguese [colleagues shared their knowledge](#) and research with fire management experts from Germany, Croatia, Greece, France,

Poland, and Italy. This was a great opportunity for WKR beneficiaries to have the opportunity to learn from fellow-European countries – especially those more advanced in coping with wildfires. This Nemausus Study Tour was originally a 'closed' project activity for consortium partners. Through valuable networks and collaboration with other projects (i.e., AFAN), WKR project in cooperation with partner Pau Costa Foundation, was able to help facilitate the participation of 7 experts from Germany and 2 from Poland and Greece.

12. Mai 2022: [AFAN Workshop – Aix-en-Provence](#)

Between April 19 and 22, experts from seven European countries participated in a very productive workshop, field exercises and final dissemination event in Aix-en-Provence, France. The activities took place in the frame of the European Union Civil Protection Mechanism (DG-ECHO) [Knowledge Network](#) project for the [Advanced Fire Analyst Network \(AFAN\)](#) and was hosted by the [French Academy for Civil Protection VALABRE](#). The European Forest Institute (Alex and Lindon) are both registered in the AFAN network ([see here for an interactive map](#)). AFAN and WRK also supported the participation of the East Bavarian Technical University [Regensburg](#), which is working closely with the Bavaria State Fire Academy Regensburg and the Bavarian Ministry of Interior to develop wildfire management capacity for the state of Bavaria, which has also enlisted the expertise of WKR.

2. Mai 2022: [Between Theory and Reality Check: PyroLife trainings in Cyprus](#)

Between 8-14 April, the EU funded [PyroLife](#) project held two training events on the island of Cyprus. This blog post shares what we, as Early Stage Researchers, did and learnt during the *Risk Communication workshop* and the in-field module of the *Making Change in Wildfire Management: Science Policy Interaction* training.

April 21, 2022: [Integrated forest fire exercise in the Harz Mountains: Forest fire expert advisor](#)

In a two-day exercise between March 25 and 27, participants from various federal states, various institutions, forestry departments, and firefighters were able to jointly practice forest fire emergency response in difficult terrain. The Harz Mountains are experiencing widespread dieback of spruce forests, thus facing an increased risk of forest fires.

5 April 2022: [Special exhibition on forest fires: WKR at the KWF Theme Days](#)

WKR Project, together with partners, is serving the topic of forest fires. As an exhibitor and co-organizer of the topic, the WKR Project, together with its partner [Forest Research Institute Baden-Württemberg \(FVA\)](#), as well as with project partners from industry, [research and practice on site](#).

March 16, 2022: [SNEP in Poland - WKR Project partner promotes proactive fire management](#)

Dear Wildfire community, it is time to officially present you **SNEP** (polish: Stowarzyszenie Niezależnych Ekspertów Pożarnictwa), which means: Association of Independent Fire Experts. The Association operates in Poland in the spirit of international cooperation and exchange to promote integrated fire (and forest!) management.

March 3, 2022: [Putting knowledge into practice - First WKR training module published!](#)

The following chapter provides a clear explanation of fire behavior, i.e., how fuel ignites, how a flame develops, and how the fire spreads. It also describes the fuel, the "food" of the fire.

January 25, 2022: [Perspective on Forest Fire Management - Interview with Elke Urbansky](#)

Elke Urbansky is part of the team at the Ostheide Forest Education Center (WPZ) of the Lower Saxony State Forests. The WPZ offers diverse programs for children and young people on the topic of forests. She is a forester, certified forest educator, and certified nature and landscape guide.

As co-founder of a forest kindergarten, she is particularly passionate about children's early childhood interaction with nature and the forest.

4. November 2021: [Wildfire Diaries Episode VII: "The living proof of the value of exchange"](#)

The latest edition of the "Wildfire Diaries" by [Vallfirest](#) with [Alex Held](#), Senior Expert of Fire Management, Silviculture and Wildlife at the European Forest Institute (EFI) and WKR project manager.

4. Oktober 2021: [Exchange of Experts in Portugal, August 2021](#)

Nico Semsch (Fire Chief of the Zehdenick Volunteer Fire Department, 33 years old) from the forest fire team participated in an expert exchange in Portugal between August 23 and September 5 and reports on his experiences. The two-week exchange in Vila Nova de Poiares, Portugal, made possible by the WKR project, focused on intra-European knowledge transfer in practical firefighting in extreme weather conditions and tactical forest fire prevention.

September 13, 2021: [Fire front Southern Europe](#)

What can we learn from the southern European countries? This Swiss film shows how Portugal and Spain are preventing forest fires. The WKR project provided input prior to the film's production.

Delivered. Protagonists from the film, including Carlos Trindade, Civil Protection Mafra, Portugal, work with fire to protect themselves from fire.

30 August 2021: [Into the future with the leaf blower](#)

The WKR project works closely with stakeholders from academia, politics, and practice. The Zehdenick Fire Department established a unit to implement training modules in local fire departments that have long proven effective in practice, especially in Mediterranean regions.

July 29, 2021: [Perspective on forest fire management - Interview with Prof. Dr. Michael Müller.](#)

Professor Dr. Michael Müller of the Dresden University of Technology's Faculty of Environmental Sciences has been teaching and researching forest fires for a long time. His research areas in forest science and forest protection also include topics related to insects, mice, and wildlife. He is, among other things, the leader of a subproject in the THOR joint project, which is also funded by the Forest Climate Fund and focuses on forest fire prevention through the management of forest structures.

July 14, 2021: [Tragedy Fires- The Human Factor: International wildland fire safety summit.](#)

At the 16th International Wildland Fire Safety Summit on May 25, Lindon was invited to give a presentation on the "human factor" in the context of fatal wildland fire operations.

July 9, 2021: [Forestry and forest fires: Results of a nationwide survey among forestry practitioners.](#)

In many federal states, forest fires are a relatively new phenomenon, although northeastern Germany has been confronted with the issue for decades. Accordingly, the situation in individual states varies considerably in terms of the density of regulations for dealing with forest fire risks.

In light of this, the Forest Climate Fund's Forest Fire Climate Resilience project investigated the extent to which forest fire prevention and preparedness for potential forest fires have found their way into forestry practice. A nationwide online survey was conducted for this purpose.

July 1, 2021: [Forest fire preparedness with the fire department - work continues on the demonstration area near Gartow.](#)

Working in a good mood and in a friendly atmosphere is easier. The work on the demonstration area in Gartow confirmed this once again. As part of the WKR project, six demonstration areas were set up in different German states. (More about the demonstration areas [here](#)) The work at Gartow ([Gräflich Bernstorff'sche Betriebe](#)) continued between 23-27 June.

14. Juni 2021: [Prescribed Fire TRaining EXchange - Cal-TREX Yuba \(Nordkalifornien\)](#)

Successful exchange in the area of "Prescribed Fire": Preparation for transfer of the TREX concept for Germany within the framework of WKR.

April 29, 2021: [Blickwinkel forest fire management interview with Detlef Maushake.](#)

Forest fire management is an integral task, which means that actors from very different areas work together on holistic forest fire management, which includes not only the actual fire extinguishing but also preventive work to increase the resilience of forests and thus minimize the negative effects of fires.

April 6, 2021: [Fighting ground fires – more than just hot air?](#)

Water is widely known as an effective extinguishing agent for vegetation fires, and considerable effort is put into transporting sufficient amounts of water to the scene quickly and efficiently. Less well-known are methods of firefighting by [removing fuel](#). For example, by creating clearings and clearing strips. Almost unknown, at least in Central Europe, is the use of [air](#), or more specifically, powerful leaf blowers, preferably backpack-mounted.

March 29, 2021: [The fire extinguishing backpack - Another tool in the toolbox!](#)

As part of the WKR project, additional equipment was delivered to the [Hardtwald demonstration area](#) over the weekend. [handed over and initial instructions](#) on handling and application are given.

In addition, a 600-liter flexible water tank (ST600) was handed over.

February 19, 2021: [Work on the Hardtwald demonstration site has begun.](#)

The WKR project has begun with the establishment of the first demonstration site. The Hardtwald is a historic forest area in the Upper Rhine Plain between Rastatt and Graben-Neudorf, north and south of Karlsruhe. The Hardtwald lies on relatively poor sandy soils in the floodplain of the Upper Rhine. The predominant tree species is pine.

January 22, 2021: [Ready for "Forest-Field-Farm"](#)

For decades, the "Skid Unit" (slip-on unit, Bakkie Sakkie, Type-6 unit) has proven itself worldwide as a mobile and multifunctional firefighting unit mounted on a pick-up truck. The areas of application are diverse and

Typically highly valued for use in initial attack, patrolling, firefighting, irrigation, securing and much more.

November 23, 2020: [A different look at the "Forestfire Climate Resilience" project](#)

In order to convey the goals of the "Forest Fire Climate Resilience" project in a video, it was important for me to experience the WKR project activities firsthand. Therefore, I participated in the [press conference in Bruchsal](#), and at a [fire brigade training in Munster](#) and have set myself the goal of addressing the topic of "forest fires", which has a large media presence, from a different perspective.

October 14, 2020: ["Life in the Situation" or fire deployment for vegetation fire fighting.](#)

The Forest Fire Service (WKR) and the Forest Fire Team (Waldbrandteam eV) jointly led a fire camp in Munster. Thirty-five motivated participants from various areas of the forestry sector (including private forest owners and the state forestry administration), two state fire schools, the Bundeswehr fire service, and various local fire departments, as well as members of the Forest Fire Team eV and Forest Fire Watch, spent a training weekend at the Munster (Örtze) military training area for practical training in controlled burning.

1 October 2020: [It only works together: New perspectives for forest fire management in Baden -Württemberg](#)

On the occasion of the launch of the WKR project and the establishment of a "model region for forest fire management" in Baden-Württemberg, the FVA invited media and press representatives to the [State Fire Service School](#) in Bruchsal on 25 September and then invited to the [Hardtwald](#) to present approaches for new ways of forest fire management.

August 31, 2020: [Students of Fire - The 2020 fire camp of the Waldbrandteam eV was a complete success.](#)

In Ehra-Lessien (Lower Saxony), on a former military training area, nearly 30 participants (aged 25-49) completed practical training as "Wildland Firefighter 1" in a two-day "fire camp" following previous theoretical sessions on forest fire fundamentals and fire behavior (online). The fire camp marks the conclusion of the basic training.

July 19, 2020: ["Vegetation fire" exercise in Faßberg.](#)

A joint training session of the Gifhorn Federal Police, the Lower Saxony State Police, the Army Aviation, the Fire Service of the Lower Saxony State Fire Brigade Association and the German Federal Armed Forces Fire Service recently took place at the Army Aviation Base in Faßberg.

Forest fire danger sign

The concept for developing a wildfire warning sign emerged through the collaborative efforts of the WKR project and its network partners. With initial seed funding, the project brought together experts from the fields of fire, education, graphic design, and communications to brainstorm and refine the idea. The goal was to design a sign that would effectively alert diverse audiences, including people with disabilities or impairments, to the danger of wildfires.



By engaging with stakeholders from various German states, including forestry organizations, insurance companies, and communications experts, we ensured that the sign design meets the needs of a wide range of people, is universally applicable, and can be used throughout Germany. Collaboration with the German Meteorological Service (DWD) facilitated access to real-time fire risk assessments and the Forest Fire Danger Index. In addition, the European Forest Institute and the Forest Research Institute (FVA) Baden-Württemberg provided up-to-date background information, accessible via QR codes on the signs. The joint WKR/FVA/DWD press release is available here: https://www.waldbrand-klima-resilienz.com/_files/ugd/bf6978_9722acc59f6648b493de3a5a4efc802a.pdf

And here:

https://www.dwd.de/DE/presse/pressemitteilungen/DE/2024/20240618_pm_waldbrandgefahrenplakat_news.html?nn=495078

Forest fire education

Ideally, education and awareness-raising begin in childhood. In the WKR joint project, neither the EFI nor the FVA had sufficient pedagogical expertise to implement a forest fire education program. However, in the WKR network, they certainly did. The Ostheide Forest Education Center (WPZ Ostheide) was already involved in the establishment of the WKR demonstration area in Lower Saxony (AP7), and thus, with expert input from the WKR project, a child-friendly forest fire climate curriculum was developed at the WPZ Ostheide.

The educational project at the WPZ Ostheide “**Forest Fire Climate**” was so successful that it was nominated for the **German Forest Education Prize**.

Future-oriented environmental education at the Ostheide Forest Education Center

(Oerrel) At the beginning of August 2022, the media reported 600,000 hectares of burned forest in Europe. This is double the previous record year of 2021. News like this motivates foresters to provide educational work and raise awareness about forest fire prevention, especially in the forest fire-prone northeast of Lower Saxony. The Ostheide Forest Education Center (WPZ) of the Lower Saxony State Forests, **in cooperation with the Forest Fire Climate Resilience Project (WKR) of the European Forest Institute, designed the educational project "Forest Fire Climate," and was thus nominated for the 2021 German Forest Education Award of the Protection Association** 51.

German Forest nominated.

"Knowledge about the occurrence, causes, and interactions of forest and vegetation fires is often insufficient or limited to calling the fire department in an acute fire situation," says project developer Elke Urbansky of the Ostheide Fire Protection Center. The educational project aims to raise awareness and promote understanding of integrative fire management. Forest fire protection concerns the entire population and begins by raising awareness of the issue among children.

The "Forest Fire Climate" project aims to provide children and young people with basic knowledge of the forest and demonstrate connections. It imparts knowledge about the forest, the soil, the humus layer, and the carbon cycle. The fire triangle is also used to illustrate how fire actually burns. The program combines traditional Boy Scout knowledge with the fascination of "making fire." The aim is to prevent fires by teaching responsible fire handling. Modern vegetation firefighting equipment can be tried out in an experiential way. Furthermore, the event is intended to motivate participants to become involved in society as firefighters.

"The nomination for the award is an indication of how important the topic of forest fire prevention has become in society. This shows us that our environmental education work is in tune with the times," says Elke Urbansky happily.

The educational program has been running since 2021 and is the most frequently booked offering at the WPZ. It is designed for children and young people from grade 3 and up. Fire departments can also take advantage of the program. @ Lower Saxony State Forests

Newsletters

Note: Originally, we planned to have more regular newsletters and far fewer blog posts. We realized that readers responded much better to blog posts with single, more easily communicated topics. Therefore, we adjusted our communication efforts and focused on more regular blog posts (averaging one per month) instead of continuing with newsletters.

Issue 2: July 2021: https://www.waldbrand-klima-resilienz.com/files/uqgd/bf6978_803ad9483ae84e57b89fea2c3e0e8691.pdf

Issue 1: October 2020: https://www.waldbrand-klima-resilienz.com/files/uqgd/bf6978_1d3af4d9c4fc41fe9092d45bb9331e71.pdf

WP6-relevant activities:

Date: December 18, 2023

WP: 6

Location: Online

Reason: Planning with DWD for the OR code on the new forest fire warning sign

Participants: Mr. Poseida, DWD, A Held, L Pronto

Short description: Involvement of DWD forest fire danger index in warning sign

Follow up: Forest fire warning sign published: <https://www.waldbrand-klima-resilienz.com/waldbrandwarnschild>

Date: December 11, 2023

AP: 6

Location: Online

Occasion: Forest fire warning sign, meeting of all project participants

Participants: A Held, L Pronto, E Urbansky, A Orth, Y Hengst

Date: September 9, 2023

AP: 6

Location: Online

Reason: Planning meeting for documentary series

Participants: Pronto, PCF, Scenery (Holland)

Short description: Discussion about filming locations and protagonists for a documentary series about fire and people (e.g. about WKR work)

Date: July 2023

AP: 6

Occasion: Daily media work / interviews

People involved: Held and Pronto

Date: June 2023

AP: 6

Occasion: Daily media work / interviews

People involved: Held and Pronto

Date: April 21, 2023

AP: 6

Location:

Online **Occasion:** Meeting with Financial Times

People involved: Pronto, Chief of Features, Financial Times

Short description: Interview and planning session to engage the Financial Times at WKR EoE and Forest Camp, Poland, to contribute to an article on the conditions facing countries in Europe and how emergency responders are training and preparing to meet new challenges posed by wildfires.

Follow up: FT cover story: <https://www.ft.com/content/540fa28d-9590-4ea2-99d7-453e3e2c3086>

Date: March 14, 2023

AP: 6, 7

Location: Online

Occasion: Planning Demo Day UXO Fire Management in Brandenburg

Participants: Held, LSTE Brandenburg, Dörling

Date: February 27, 2023

AP: 6

Location: Online

Occasion: Discussion with "Brennpunkt Wald"

Participants: Brennpunkt Wald, Held, Pronto, Schloßmacher

Short description: Support for the creation of training materials for Brennpunkt Wald

Follow up: Lindon makes videos

Date: 17 February 2023

AP: 6, 7, 3

Ort: Ramleh, Lebanon

Anlass: Workshop: "Support Local Action for an Efficient First Response: Structure, Roles and Responsibilities" hosted by Ministry of Environment, Lebanon

People involved: Pronto

Short description: The WKR training content was translated into Arabic and presented at a workshop with various national stakeholders on behalf of the World Bank's Technical Assistance for Sustainable Forest Management and Forest Fire Prevention in support of the Lebanese Ministry of Environment.

Follow up: Possibility for a follow-up project to translate the WKR modules into Arabic for the Lebanese context.

Date: November 29, 2022

AP: 6, 7

Location: Online

Occasion: Planning meeting with the International Association of Wildland Fire

Participants: L Pronto, A Held, M. Robinson and C. Bushy (IAWF)

Short description: A meeting was held to discuss the possibility of Germany hosting the "7th. International Conference on Fire Behavior and Fuels."

Follow up: examines the interest and feasibility of an international fire service conference in

Germany in 2024

Date: November 10, 2022 AP: 6, 3

Location:

Hybrid / Black Forest **Event:**

Virtual Fires Conference (podium discussion) - St. Georgener Technologiezentrum GmbH **Participants:** L. Pronto **Short**

description: Lindon participated as a

guest speaker in a panel discussion at the Future Fires Congress on the topic "Our world in danger: wildfire detection from space using satellites and artificial intelligence".

Date: 27 September 2022 AP: 6, 7

Location:

Online

Occasion: Planning meeting with International Association of Wildland Fire **Participants:**

L Pronto, A Held, M. Robinson and C. Bushy (IAWF)

Short description: A meeting was held to discuss the possibility of Germany hosting the "7th. International Conference on Fire Behavior and Fuels.

Follow-up: examines the interest and feasibility of an international fire service conference in Germany in 2024

Date: August 8, 2022 AP: 6

Location: Munich

Occasion: Evening Show Studio Guest at Bayerischer Rundfunk

Participating Persons: L. Pronto

Date: 20-24 June 2022 AP: 6, 3

Location:

Hanover

Event: INTERSCHUTZ Trade Fair

Participants: L. Pronto, A. Held **Short description:**

Numerous networking activities and meetings with various international and national technology partners and organizations

Date: April 8-9, 2022 (10-13)

AP: 6

Location: Cyprus

Occasion: Risk Comms Workshop Part 2

Participants: Alex, Maria and Ewa

Short description: PyroLife comms workshop in Nicosia, continuation of the online course in November

Date: April 4, 2022

AP: 6

Location: Online

Occasion: Pyro-Life PhD Training Event: Making Change in Wildfire Management: Science Policy Interaction **Participants:** L. Pronto

Date: March 31st - April 2nd, 2022 AP:

6, 3

Location: Jessen, Saxony-Anhalt

Occasion: KWF theme days with WKR-organized special exhibition on forest fires **Participants:**

Lindon, Alex, Maria **Short description:** Efi/WKR as

co-organizer and exhibitor at the KwF theme days with a large, approximately 80 meter long stand Also present: Vallfirest, FVA, Euro-Waldbrand

Follow up: "Heizpilz" with Bosch, follow up in June and at various events

Date: 08 March 2022

AP: 6, 3

Location: Online

Occasion: WebTalk Series: Climate Change and Disaster Management WebTalk #3: Forest Fires

Participants: L. Pronto; Bonn Network for International Disaster Protection and Risk Management; Fraunhofer INT; Forest Europe; WKR, DKKV, FW /City of Bonn, Fraunhofer, DHL, UNU, University of Bonn, Welthungerhilfe, City of Cologne, Germany Helps.

Short description: Fraunhofer INT reports on the Firelogue project to improve forest fire risk management. Forest Europe presents its current work toward a cross-border advisory mechanism in the area of forest damage, using forest fires as an example. The European Forest Institute (EFI) also reports on its current thematic work, particularly the Forest Fire Climate Resilience project, which focuses on improving forest fire management in Germany.

Follow up: Founding of the Bonn Network for International Disaster Protection and Risk Management AG Waldrand.

Date: March 4, 2022

AP: 6

Ort: Online hybrid/ Claremont CA, USA)

Anlass: Roberts Environmental Center (Claremont McKenna College, Claremont CA, USA) 8th Annual Green Careers Conference

People involved: L. Pronto

Brief description: Advocacy Panel - Discussion on wildfire experiences and examples from around the world, including the WKR project: <https://rec.cmc.edu/gcc2022/>

Date: February 21, 2022

AP: 6, 7

Location: Jessen

Occasion: Working meeting to plan KWF theme days in Jessen

Participants: L. Pronto, A. Held

Date: 12 January 2022

AP: 6

Location: Online

Occasion: Expert Interview

Participants: L. Pronto; Rosenbauer Research Department

Short description: Discussion about coordination in forest fire operations, challenges/opportunities/needs in Germany and Austria

Date: November 2021

AP: 6

Location: Online

Occasion: Comms Workshop

Participants: Maria (support facilitation and organization)

Short description: PyroLife

Date: 29 September 2021

AP: 6

Location: online

Occasion: Forest education workshop Elke Urbansky with Michael Herrmann

Participants: Elke Urbansky, Micha Herrmann, Maria, about 45 participants

Short description: Presentation of the WKR project by Maria (briefly) following Micha's presentation.

Date: 24-27. May 2021

AP: 6, 3

Location: Online

Anlass: Wildland Fire Safety Summit/ Human Dimensions Conference (Keynote: "Human and Environmental Factors on Tragedy Fires)

Participants: several hundred participants worldwide (online)

Kurze Beschreibung: Keynote: "Human and Environmental Factors on Tragedy Fires

Date: November 6, 2020

AP: 6

Location: Online

Anlass: Presentations to Angell Students

People involved: L. Pronto

Short description: Two presentations for students in grades 10 and 11 about the

Fire management in the USA and the forest fire situation in Germany and the WKR project were presented.

Follow up: <https://www.angell-montessori.de/allgemeines/alle-neuigkeiten/allgemein/virtuelles-treffen-mit-einem-us-firefighter>

Date: 27 October 2020

AP: 6

Location: Online / Bonn

Occasion: WKR presentation to EFI Bonn Office (Scientific Seminar)

Participants: L. Pronto, A. Held, M. Schloßmacher; All staff of Bonn office

Short description: Presentation to the project to EFI Staff

Date: 25 September 2020 **AP:** 6, 3

Location: Bruchsal

Occasion: Press conference model region

Participants: EFI-Lindon, Alex, Rosa, Maria; FVA-Christoph, Yvonne, Carolin, as well as various press representatives and project partners, e.g. Hagen Vollrath (THW)

Short description Follow

up: <https://www.waldbrand-klima-resilienz.com/post/es-geht-nur-gemeinsam-neue-perspektiven-f%C3%BCr-waldbrandmanagement-in-baden-w%C3%BCrttemberg>

Date: 11-12 September 2020

AP: 6, 3

Ort: Münster FireExpo

Occasion: Fire Department Exhibition FireExpo

Persons involved: Held, KLP

Short description Held as a specialist speaker at the FireExpo, and presentation of Robogreen Firemaster prototype

Date: 27. July 2020

AP: 6

Location: EFI Bonn

Occasion: Meeting with Phoenix journalist Mr. Fittkau

Participants: Mr. Fittkau, Alex, Maria

Short description: Mr. Fittkau received the PM about WKR after I saw one of his forestry programs on Phoenix. He then contacted me, saying he was interested in producing something on the topic (next year). The first meeting went well; he was very interested.

Follow up: Definitely invite him to this fire drill for journalists next year (planned for spring 2021)

AP 7 Demonstration areas and training, workshops

The "Forest Fire Climate Resilience" project aims to develop and test strategies to increase the resilience of forests to wildfires and the impacts of climate change.

Demonstration areas play a central role in practically testing, training and evaluating various measures.

Role, function and application of the results of the demonstration areas

Increasing forest fire resilience: Various methods for

Increasing resilience to forest fires and climate impacts is being tested. This includes various management strategies such as selective timber harvesting, the promotion of specific tree species, and the creation of firebreaks, buffer zones, and fire lanes.

Biodiversity and species conservation: Another goal is the synergistic promotion of biodiversity. The areas are being studied to determine how different measures influence the occurrence of flora and fauna. Rare or endangered species also play a role.

Monitoring and data analysis: The areas are also used for scientific observation and

Data collection. Sensors and regular site visits help document the impacts of the measures on soil, vegetation, and microclimate. This data is essential for further developing and adapting strategies.

Training and further education: All demonstration areas served and continue to serve the active training on 56

Object. The land is continuously used by the owners as well as forestry training centers and fire schools.

Value of demonstration areas

The demonstration plots have high scientific and practical value because they serve as living laboratories. They allow for real-life experiences of how various strategies to increase forest fire resilience can be implemented. This is important not only for science, but also for practical forestry and nature conservation.

Scientific findings: The collected data and experiences help to improve the understanding of to deepen the impacts of climate change on forest ecosystems and to develop new models and forecasts.

Practical solutions: Forestry and nature conservation benefit directly from the findings, as they are prepared in the form of recommendations and guidelines for practice.

Education and awareness: The demonstration areas will also be used for educational purposes to To inform professionals, students and the public about sustainable forest management practices and how to deal with the impacts of climate change.

Location and location of the areas

The demonstration plots are located in different regions, representing different forest types, ownership structures, and climate zones. This diversity allows for a wide range of conditions to be studied and the results to be extrapolated to other regions.

The exact geographical coordinates, elevations, soil composition, and existing vegetation types vary depending on the location. This diversity is important for testing the effectiveness of measures under different conditions and drawing general conclusions.

Work on the demonstration areas will continue after the end of the project, including in Brandenburg, Lower Saxony, Bavaria and Baden-Württemberg.

The FNR Media Day 2022, a public exercise at the WKR demonstration area in Lower Saxony, was a complete success.

Demo region Bavaria "Chiemgau Alps"

Between 2021 and 2024, several actions and training activities on the topic of *slashing* and follow-up firefighting were carried out in the "Chiemgau Alps" demonstration region together with mountain farmers, mountain rescue services, and local fire departments. WKR was also invited to conduct forest fire awareness and early childhood safety training at a forest kindergarten in Grassau as part of a demonstration region activity. In a further development of the collaboration with the Bavarian Mountain Rescue Service, a training session was also conducted with the Füssen Mountain Rescue Service and the Schwangau Volunteer Fire Department. The focus was on a forest fire drill in difficult terrain—this time with the iconic Neuschwanstein Castle and the Bavarian Alps as a backdrop. As forest fires become increasingly common in the Alpine region, it is important to be appropriately prepared for firefighting in complex environments—dangerous terrain, tourists, cultural heritage sites—this requires the collaboration of various stakeholders, such as firefighters supported by the Mountain Rescue Service, and awareness-raising through public education.

Selected press reports:

- Newspaper article: July 23, 2023, **Süddeutsche Zeitung** (online and print): "Fighting the flames with a hoe and helicopter" (Pronto)
- Newspaper article: August 28, 2022, **OVB - Oberbayerisches Volksblatt** (online and print): "Amidst blazing flames: How a firefighter from Aschau im Chiemgau fights forest fires" (Pronto)
- TV interview: Evening news segment: July 21, 2023, **Bayerischer Rundfunk**: (Abendschau-South) (Pronto)
- TV report: Joint exercise for forest fire prevention: July 24, 2024, **Bayerischer Rundfunk**: (We

in Bavaria) (Pronto)

- Newspaper article: July 18, 2024, **Allgäuer Zeitung** (online and print): "With a view of Neuschwanstein: Füssen Mountain Rescue and Schwangau Fire Department Fight Flames" (Pronto)
- Press release: July 15, 2024, **Bavarian Mountain Rescue Service**, Füssen "Forest fire on camera" (Pronto)



Forest fire drill near Neuschwanstein Castle



Handing over tools to Chiemgau alpine farmers/herdsmen and mountain rescue service with Aschau/Sachrang fire department and Aschau mayor Simon Frank.



Forest fire awareness at the Grassau forest kindergarten



One of the numerous “Schwendaktionen” and training activities in the Chiemgau Alps demo region

Demo area Lower Saxony

The demonstration area in Lower Saxony (see <https://www.waldbrand-klima-resilienz.com/post/waldbrandvorsorge-mit-der-feuerwehr-arbeiten-an-der-demo-fl%C3%A4che-bei-gartow-gehen-weiter>) is located in a large private forest and on and near the burnt areas of the catastrophic forest fire of 1975. In cooperation with the network partners, the areas were established, maintained, and used for excursions, training, and continuing education. (See also Annex.) The experiences from 1975 raised awareness of forest fire prevention very high, and both the forest owner and the surrounding communities and fire departments were correspondingly motivated. In addition to fire load reduction along a country road, silvicultural measures were implemented, such as reducing the number of trunks, planting field maple understory, and reforesting the entire area. Examples include:

For excursions, seminars and active work on and with the areas, the FNR media day on the demonstration area in Gartow can be mentioned here: <https://www.kiwuh.de/presse/pressemitteilungen/aktuelle-nachricht/waldbrandgefahr-365-tage-im-jahr-managen>

In addition to silvicultural and technical measures, active fire work was also possible here, as the forestry operation also includes open-land nature conservation areas that are maintained with fire. This enabled training in the use of fire as a tool for forest owners and the surrounding fire departments.



Networking event at the demo area in Gartow, forestry participants from Lower Saxony Baden-Württemberg and Mecklenburg-Western Pomerania, private state and municipal forests, THW and fire brigades



FNR media day at the demonstration area in Gartow with tactical fire support and involvement of regional forces for training purposes.

Another result of the demonstration area in Gartow is an educational film produced with local partners:

Forest fire management is a cross-sectional task that fire departments cannot solve alone.

As part of the project Forest Fire Climate Resilience (WKR Project) funded by the German Federal Government and based at the European Forest Institute (EFI), this film was produced in cooperation with Dr. Michael Herrmann and the Ostheide Forest Education Centre of the Lower Saxony Forestry Commission.

The report was produced by the State Forestry Office (Elke Urbansky). It is intended to encourage efforts to develop regional solutions for improved forest fire prevention. It is undisputed that climate change is leading to an increasing predisposition of forests to forest fires.

The aim of the film is to contextualize the knowledge and lessons learned from 1975 and translate them into practice. It is aimed at all stakeholders in forest fire management and all types of forest ownership. The film is freely available, and public viewing is expressly encouraged.

The film is available here: https://youtu.be/ezoQS_KMR6o?si=Ce6XU4Bt0ks4RiEh



Selected press reports:

- Interview: December 1 (2023), **Wendland-net**: Training: "So that the forest does not go up in smoke" (Held)
- Interview/Video: February 1 (2023), **FNR**: Forest Fire Prevention (Held) • Interview/Video: February 1 (2023), **FNR**: Is Deadwood Fuel? (Held) • Interview/Video: February 1 (2023), **FNR**: Modern Forest Fire Management (Held)
- Interview: January 4 (2023), **AFZ The Forest**: Climate change requires vision in fire management (Held)
- Interview: October 11, 2022, **FNR**: "Managing forest fire risks all year round" (Held)
- TV interview: September 24, 2022, **Elbe-Jeetzel-Zeitung**: "At Trebel: Fighting fire with fire" (Hero and Pronto)
- TV interview: 26 September 2022, **Sat.1** regional: "Heide in Flames: Large fire brigade exercise in Lüchow" (hero)
- Interview: June 29, 2021, **EJZ**: "Gartow Forest: Depriving the forest fire of fuel" (Held)
- TV interview: June 24, 2021, **NDR**: "Fuel buffer zones to prevent forest fires" (Held)
- Interview: 4 May 2021, **STATE FORESTRY**: "Cooperation in forest fire prevention" (Held) • Radio: 3 May 2021, **ANTENNE**: "Forestry office receives firefighting tanks for pick-up trucks: Forests to dry" (hero)

Demo areas in Baden-Württemberg

Bruchsal

A demonstration area for integrated forest fire management was established in cooperation with the Hardtwald Forestry Company (Forst BW) in the immediate vicinity of the Baden-Württemberg State Fire Service Training School. Both the Forestry Company and the State Fire Service Training School used and continue to use the area for their in-house training and continuing education courses.

Based on the demonstration area and inspired by the WKR project, the FVA partner came up with the idea of establishing a model forest fire region for Baden-Württemberg in the Rhine-Neckar climatic hotspot, based on similar model regions for storms and beetles. In this context, a so-called tandem forest fire concept for Baden-Württemberg was also developed.

The tandem describes the professional cooperation between forestry and fire service.



Model region:

The "Model Region Forest Fire" in Baden-Württemberg is an innovative initiative that aims to Forest fires through a combination of prevention, early detection and efficient

To better manage firefighting measures. The model region was established as part of the WKR project to develop and implement a comprehensive strategy for managing forest fire risks.



Development history

Initiation and needs identification

In recent years, the frequency and intensity of forest fires have increased in Europe, including Germany. This is due to climate change, increasing dry periods, and increased fire danger. Baden-Württemberg recognized the need to respond to this challenge early on and therefore began to implement systematic measures for forest fire prevention and

to develop a fight against it.

Start of the WKR project

The WKR project was launched to create a platform for the exchange of knowledge and experience in forest fire management. It brings together various stakeholders, including forestry authorities, fire services, scientists, and policymakers. The goal is to

to develop and test innovative techniques and strategies that can be applied in the model region.

Implementation of the model region

The "Model Forest Fire Region" was established in specific areas of Baden-Württemberg that are particularly vulnerable to forest fires. These regions were carefully selected to provide a representative base for the development and testing of measures. Data is continuously collected in these areas and used for research and development of new technologies and methods.

Core components of the model region

Preventive measures

One focus of the model region is prevention. This includes regular inspections and maintenance work to reduce the fire load in forests. These include measures such as removing undergrowth, installing fire barriers, and monitoring forest areas.

Early detection systems

Another important aspect is the implementation of early warning systems. Modern technologies such as drones, satellite monitoring, and gas sensors are being used to detect early signs of wildfires. These systems are capable of identifying temperature changes, smoke, and other indicators and triggering an alarm before a fire can spread.

Efficient firefighting

In the event of a forest fire breaking out, the model region has developed special operational plans to ensure rapid and effective firefighting. These include the deployment of specialized emergency personnel, the use of firefighting aircraft and helicopters, and the use of modern firefighting techniques.

Cooperation and further training

A key component of the project is collaboration between different institutions and the continuous training of the participating actors. Regular training sessions, workshops, and exercises are conducted to ensure that all participants are up to date with the latest technology and tactics.

Outlook and further development

The findings and experiences from the "Model Region Forest Fire" will serve as a basis for further developing and expanding the measures to other regions. In the long term, the project aims to contribute to establishing a robust and effective system for forest fire prevention and control throughout Germany and beyond.

The model region in Baden-Württemberg is already demonstrating how targeted measures and innovative approaches can minimize the impact of forest fires and offers an example of how other regions can prepare for similar challenges.

Tandem concept forest fire

Minister Peter Hauk MdL: "The current forest fire situation in Germany shows that we must adapt not only the forests but also our forest risk management in the country to climate change in order to be well prepared."

Deputy Prime Minister and Interior Minister Thomas Strobl: "The forestry administration and fire department are working closely together and are preparing for climate change and its consequences."

Forest fire protection / Hardtwald model region: Expert advisory tandems from the fire service and forestry have proven successful.

"Climate change is significantly increasing the risk of forest fires in Baden-Württemberg, despite well-maintained and predominantly mixed forests. Data from the German Weather Service demonstrates an annual increase in the number of days with a high risk of forest fires since 1980. This also increases the risk of vegetation fires."

That is why we must take precautions now and organize preventive measures, because successful forest fire fighting requires comprehensive forest fire management," said the Minister for Food, Rural Areas and Consumer Protection, Peter Hauk MdL, on Wednesday (12 August) in Stutensee (Karlsruhe district), on the occasion of an information event in the 'Hardtwald Model Region for Forest Fire Protection' and the presentation of initial project results and modern fire protection technology in the forest.

The Rhine plain and northern Baden-Württemberg are particularly at risk, which is why a model region for forest fire management has been established in the Hardtwald forest in the northern Rhine plain between Ettlingen and Heidelberg. Minister Hauk emphasized that procedures and concepts are being developed, tested, and practiced here in close collaboration between practitioners and researchers, including firefighters, forestry, and other stakeholders.

"The forestry administration and the fire service already work hand in hand – and we are now strengthening this collaboration even more closely with the fire service/forestry advisory tandems. In emergency response, it is essential to adapt to changes in a timely manner, and we take climate change and its consequences particularly seriously. That is why, for example, we jointly organized a statewide forest fire symposium in 2017 on my initiative. A good example of the already excellent cooperation is the state's extensive network of forest trails. This enables our fire service units to quickly fight forest fires on the ground," Deputy Prime Minister and Interior Minister Thomas Strobl encouraged the project participants.

"Together with the fire department, the State Forestry Administration and ForstBW are addressing the challenges and implementing exemplary preventive measures in the state forest," said Minister Hauk. These include, among others:

*the construction of fire water wells in consultation with the municipalities and forestry measures,
Practicing firefighting procedures with the fire brigade in the forest,
Providing digital information for the operational management and
Forest fire risk maps currently under development.*

The Baden-Württemberg Forest Research Institute is coordinating the model project and the network work and is responsible for knowledge transfer at the state level in the 'Platform for Forest Fire Management BW', which is being jointly established between the Interior and Forestry Administration.

"Based on the testing in the model region, I think it makes a lot of sense to introduce the concept of fire service/forestry advisory tandems in other parts of the state," said State Fire Director Thomas Egelhaaf.

Numerous tasks still need to be addressed by the 'Forest Fire Management Platform BW', including questions regarding specific technical equipment for fire departments and forestry departments.

@ForstBW

Forest fire management: Minister Peter Hauk MdL: *"As part of the forest strategy, the Forest Fire Platform Office supports the state's stakeholders in establishing a modern forest fire management system." Establishment of the Forest Fire Platform Office / Completion of the 'Forest Fire Climate Resilience' project in the Hardtwald model region*

"The consequences of climate change require us to realign our crisis management and adapt it to the current conditions. The increase in dry periods unfortunately also means an increased risk of forest fires in Baden-Württemberg. Data from the German Weather Service show that the number of days with a high risk of forest fires has been steadily increasing since 1980. For decades, we have been developing our forests into mixed forests with a predominance of deciduous trees.

increases the resilience of the forest ecosystem and simultaneously reduces the risk of forest fires. In addition, we have further developed the interaction between forest managers and firefighters in the prevention and fighting of forest fires in the Hardtwald model region. With the tandem concept of forestry and firefighting, we are achieving a new dimension in forest fire management in Baden-Württemberg, which will serve as a model for the state and beyond. This means that we are well prepared for the new forest fire season," said Peter Hauk MdL, Minister for Food, Rural Areas and Consumer Protection (MLR), on the occasion of the forest fire exercise marking the conclusion of the 'Forest Fire Climate Resilience' project in the Hardtwald model region, which took place on Saturday (April 22) in Kronau (Karlsruhe district).

In order to be well prepared for future forest fires, around 250 members of the fire departments from the Karlsruhe district and the Rhein-Neckar district will be practicing emergency situations in the Lußhardtwald this Saturday together with the forestry authorities and other stakeholders.

Tandem concept of forestry and fire service

In the Hardtwald model region, foresters have developed exemplary silvicultural methods in recent years to make the forest more resistant to fire. At the same time, cooperation between the forestry administration and the fire department has intensified.

*"With the **tandem concept** of forestry and fire service, we are implementing a goal of the coalition agreement and thus creating a powerful tool for modern forest fire management in Baden-Württemberg. The dynamic development of forest fires requires smooth communication and coordination between stakeholders and emergency services so they can respond quickly to changing conditions and operational situations. Other districts can now build on these findings and update their forest fire management," said the minister.*

Forest fire maps

Another aspect of the cooperation between forestry authorities and fire departments is the provision of forest fire response maps. "I am very pleased that we have succeeded in launching a forest fire response map according to uniform federal standards. The State Forestry Administration created these maps and made them available to the fire departments via the lower forestry authorities," said Minister Hauk.

Office of the Forest Fire Platform

In the Hardtwald model region, the 'Forest Fire Climate Resilience' project is coming to an end. To further expand on the findings, an office for the Forest Fire Platform is being established at the Freiburg Forest Research Institute (FVA) under the leadership of the Freiburg Regional Council. It is part of the Forest Strategy and is funded from it. "This platform continues the intensive exchange between stakeholders and supports them in establishing modern forest fire management. I am very pleased that we have succeeded in allocating the necessary funds for this in the budget as part of the Forest Strategy. This is an investment in preserving our forests and protecting the population from forest fires," explains Minister Hauk. @ForstBW

Demo area Hartheim

In the Hartheim am Rhein municipal forest, measures for forest fire prevention and fire service deployment preparation are being implemented in close coordination with the Baden-Württemberg Forestry Association (FVA) and the forest district manager (TS) and the senior forester (HA). HA has participated in several WKR training courses on the topic and has been able to gather input from various federal states. TS is writing his trainee thesis on the implementation of these measures, which he accompanies and supports with the WKR initiative.

Specifically, the municipality has purchased a small mobile forestry mulcher, portable pumps, and hand tools. Brush cutters are also being used. Buffer zones with reduced fire load are being established along strategically important routes to reduce fire intensity and enable firefighters to more effectively fight the fire. The continuity and quantity of fuel, as well as the vertical and horizontal arrangement of fuel, are being reduced. To avoid creating "fire acceleration zones," sufficient canopy cover is being ensured over the buffer zones.

The newly created park-like structures create synergies with the areas of tourism and recreation through a pleasant forest landscape, and with nature conservation through more light and more open structures, the promotion of orchid species, etc. And new opportunities also arise for hunting operations, such as driven hunts, the sowing of small grazing areas, etc.

The next steps will be excursions with the fire brigade, local council, nature conservation, but also the school and forest kindergarten to explain the effectiveness of the measures and to discuss, plan and then practice possible deployment options together with the fire brigade.

Selected press reports:

- TV documentary: August 1, 2024, **SWR STORY**: Risk of forest fire – How good are our fire departments prepared (Hero, Pronto)
- Radio interview: March 21 (2023), **SWR**: How Karlsruhe prepared for forest fires (Held) • Interview: February 15 (2023), **Oberoerdingen Volunteer Fire Department**: Participation in the colloquium of the Baden-Württemberg Forest Research Institute on the topic of "forest fires" (Held) • Interview: August 10, 2022, **Badische Neueste Nachrichten**: "Agriculture Minister Peter Hauk learns about forest fire fighting in Stutensee" (Held) • TV interview: July 26, 2022, **ZDF-Drehscheibe**: "The risk of forest fires is increasing – in the model region"

In Baden-Württemberg there are concepts to protect the forest" (Held)

- TV interview: September 28, 2020, **SWR**: "Forest fire prevention: model project in Baden-Württemberg" (Held)
- Interview: 25 September 2020, **Süddeutsche Zeitung**: "Baden-Württemberg prepares for forest fires" (Held)

Demonstration areas Brandenburg**NLB Foundation**

The Brandenburg Natural Landscapes Foundation (NLB) is one of the largest landowners in Brandenburg.

At the same time, the foundation's land is highly prone to fire, contaminated with military waste and munitions (Unexploded Ordnance, UXO). The land also has one of the highest forest fire rates in Brandenburg.

The foundation is an active partner in fire management and has been actively investing in preventive measures since 2013.



WKR-NLB forest fire protection

The Brandenburg Natural Landscapes Foundation implements coordinated forest fire protection concepts on its land. The goal is to increase safety in the event of forest fires, protect surrounding areas from fire spread, and keep the munitions-contaminated wilderness areas in the core zone of the nature reserve free from encroachment.

The forest fire protection systems implemented by the Brandenburg Natural Landscapes Foundation in cooperation with districts, fire departments, forestry administration, nature conservation, and other experts are regularly maintained and tested during fire drills. Fire incidents are evaluated jointly with all partners, new needs are adapted to the current situation, and measures are implemented with the help of requested funding and the foundation's own resources.

The foundation is a partner in **the Forest Fire Climate Resilience joint project, promoting professional exchange, training, and qualification in forest fire fighting and prevention.** The project is being implemented by the European Forest Institute and the Baden-Württemberg Forest Research Institute, together with numerous regional and international partners, with funding from the Forest Climate Fund.



The WKR project was able to provide operational support during a major forest fire on the Brandenburg-Saxony border in 2022. Through networking and trust-building within the WKR project, the three specialized German forest firefighting teams worked together for the first time.

Measures during the WKR project period:

Jüterbog, Lieberose, Tangersdorf

- Construction and expansion of forest fire protection barrier systems, new fire protection strips with a length of 7.25 km and 3.72 hectares.
- Maintenance/care of forest fire protection systems on 317.97 hectares.
- Maintenance/care of forest fire protection systems on 120.7 hectares.
- Repair of paths over a length of 2,070 meters.
- Repair of paths over a length of 2,662 meters.
- Repair of paths over a length of 2,100 meters.
- Installation of 3 new fire water extraction points.
- New installation of 4 fire water extraction points:
- Repair of paths over a length of 5,120 m.
- New installation of 2 fire water extraction points/ new installation of wound strip "Töpperweg" 1,300 m:
- New installation of 2 fire water extraction points:

Heidehof- MB III Prevention of forest damage:

- Installation of 2 new fire water extraction points.

Jüterbog:

- Survey of explosive ordnance on 4.59 hectares and creation of forest fire strips on 2.02 Hectares.
- Jüterbog: Old Camp – Measures II Project to address the Extreme weather events caused consequences in the forest
- Survey of explosive ordnance on approx. 1.46 hectares and creation of forest fire strips on approx. 1,35 km.

Hatzfeld:

With the Hatzfeldt-Wildenburg Forestry Administration, the Forestry Commission (WKR) has gained a cooperation partner from a large private forest. The Massow branch of the operation is located south of Berlin and is characterized by traditional pine management. The forest is being actively restructured according to ANW (Working Group for Natural Forest Management) guidelines. The risk of forest fires is typically very high in this area. Within the framework of the WKR project, this demonstration area served to illustrate the conceptual, long-term forest fire prevention through forest conversion. The WKR project provided advice, provided equipment, and developed the project with the forestry company.

Field trip guide on the topic of forest fire prevention. The demonstration area was also used for the field trip of the 2023 ANW Annual Conference on this topic. The equipment provided by the WKR project was and continues to be used regularly.

Selected press reports:

- Interview: June 7, 2023) **AFZ-the Forest:** Against the Fire (Hero)
- TV interview: July 24, 2022, rbb 24: "How can a fire department save water during forest fire operations? Visit to the special 'Vegetation Fire' unit and WKR project partners" (Semsch)
- Interview: June 26, 2022, **Luasitzer Rundschau+**. "Forest fire on the border of Brandenburg and Saxony – support from the USA (Pronto)
- TV interview: June 25, 2022, **RBB-24 Brandenburg Aktuell**. "Forest edge under control." (Pronto)

Demo area North Rhine-Westphalia

Drover Heide

The Drover Heide is a nature reserve covering approximately 600 hectares (ha) in the Düren district of North Rhine-Westphalia. It is located in the municipalities of Kreuzau and Vettweiß. Since April 2016, the NRW Foundation has owned 636.86 hectares of land previously owned by the federal government.

From 1914 onwards, the Drover Heath served as a parade ground for the garrison in Düren, and for its recruit training during the First World War.[2] It was confiscated as a military training area in 1951 and used by the Belgian armed forces formerly stationed in Düren, as well as by German armed forces from the surrounding areas. The area was opened to the public after the withdrawal of the Belgian armed forces at the end of 2004 and can now be explored on marked paths. In total, the area was used for military purposes for about 100 years. Due to bombing and shelling at the end of the Second World War and temporary use as a military training area, the area is contaminated with munitions and therefore the protected area may not be accessed outside of paths. About 120 hectares of the total area are actual heathland, 290 hectares are woodland, and 150 hectares are fenced and are grazed by cattle and goats to keep the heath short (de-bushing). In addition to Scottish Highland cattle, goats, such as the Thuringian Forest goat, are particularly well-suited for keeping the birch, heather, and juniper trees short. The Drover Heath is part of the Natura 2000 area with site number DE-5205-301.

Since 2020, the Drover Heide has been part of the High Fens-Eifel Nature Park.[3]

The above-mentioned conditions make the heath a refuge for endangered animal and plant species.

Many Red List species occur here. To protect the heathland and prevent it from becoming woody, the areas must be regularly maintained. Habitat management measures include extensive grazing (sheep grazing) and scrub removal.

In addition to the usual bush clearing measures, biotope maintenance through fire is also being carried out in the Drover Heath. These measures, scientifically supported by the WKR project, have shown good results in practice and mimic natural fires caused by lightning strikes, for example.

The fire operations conducted by WKR served as demonstration areas for the use of fire in both prevention and tactical firefighting. To maximize the added value of conservation burning, both fire departments and members of the @fire and Waldbrandteam eV associations participated in all operations.



Training and further education in AP7 (2020-2023)

As part of WP7, the construction, installation and use of the demonstration areas, numerous training courses, seminars and excursions were also held. In addition to the activities directly led by WKR, The area was used for numerous events by the owners and the adjacent fire schools.

Online-Workshops (2020-2023):

2x NLB Foundation: "Application of Fire" and "Fundamentals of Firefighting" for NLB employees and Teltow Fläming Fire Department

2x Nature Foundation David: "Follow-up extinguishing and fire watch", employees of nature conservation foundations and Landowner

18 face-to-face workshops (2020-2023):

- **Interforst:** WKR organized a forest fire workshop with the EU project FIRE-IN (DG-Home) as side event at Interforst in Munich
- **Luckenwalde:** "Application of fire" for NLB employees, SKV Zehdenick, Teltow-Fläming fire department
- **Nideggen:** Training on basic forest fires for the Nideggen fire department
- **Bergen:** "Training Basics of Forest Fire and Application of Fire" for the German Federal Armed Forces Fire Service
- **Munster:** "Training Basics of Forest Fire and Use of Fire" for the Bundeswehr Fire Brigade
- **3x Gartow:** "Forest fire prevention through forestry and the use of fire" for landowners and surrounding fire departments
- **Goslar:** "Forest fire specialist training and exercise" with Waldbrandteam eV
- **3x Karlsruhe:** "Training Basics of Forest Fire" at the Forestry Training Center FBZ
- **Bruchsal:** "Forest fire prevention training" for the ForstBW trainees
- **Mattenhof:** "Forest fire prevention training" for the ForstBW trainees
- **Furtwangen:** Lecture on the basics of forest fires at Furtwangen University
- **3x Freiburg:** Lecture on the basics of forest fire at the University of Freiburg, Institute of Silviculture

In addition, WKR participated in or organized several other workshops and trainings together with partner organizations and project partners not listed here.

WP7-relevant activities:

Date: October 1-3, 2023

AP: 7, 3, 6

Ort: Landsberg, Bayern

Occasion: OroraTech International Forest Fire Conference

Participants: Held, Pronto, 60 participants (exclusive)

Short description: Pronto and Held gave lectures and presented the success of WKR

Follow up: Repeat in 2024

Date: July 19, 2023

AP: 7, 6, 3, 5

Location: Aschau, Bavaria

Occasion: Handing over hand tools to alpine farmers, Mountain Rescue: Chiemgau Alps Demo **Participants:** L.

Pronto, Mountain Rescue Bavaria, FWF Aschau, FWF Sachrang, Aschau Mayor

Short description: WKR could procure tool kits for 8 mountain huts to enable safe traditional burning and rapid response to forest fires in remote areas.

Date: July 19, 2023

AP: 7, 6, 3

Location: Sachrang, Bayern

Occasion: Chiemgau Alps demonstration and educational work

Persons involved: L. Pronto, Bavarian Mountain Rescue Service, FWF Aschau, FWF Sachrang, District Fire Inspector

Rosenheim District, Rosenheim District Office, Süddeutsche Zeitung, Bayerischer Rundfunk

Short description: The WKR and mountain rescue team's collaboration is presented, and a demonstration is presented for the media. <https://www.sueddeutsche.de/bayern/waldbrand-bayern-gebirge-feuerwehr-geld-1.6057175>
https://www.br.de/br-fernsehen/sendungen/abendschau-der-sueden/abendschau-brandbekaempfung-100.html?fbclid=IwAR3NSBAFCH_e_wgwrPUWsYrdS882tS052q-vZPi3QIZvNnYH4ZdengWTCXk

Date: July 15, 2023

AP: 7

Location: Chiemgau Alps (Geigelstein)

Occasion: Swathing and training of alpine farmers in fire extinguishing work

Participants: L. Pronto, Alpine farmers

Date: 08-10. May 2023

AP: 7, 6, 8

Ort: Gartow

Occasion: FNR excursion in Gartow demo area

People involved: Held, Pronto, FNR, 100+ people

Short description:

https://wald.fnr.de/fileadmin/Projekte/2023/Artikel/Plothe_Reportage_Waldbrand_2022_ste.pdf

<https://www.youtube.com/watch?v=M9SXIn4tDpE&list=PLyeUudrNoHk5981V3gOJLdmgFc4M7cfX-&index=2>

https://www.youtube.com/watch?v=5_Ff1IO58qQ&list=PLyeUudrNoHk5981V3gOJLdmgFc4M7cfX-&index=1

<https://www.youtube.com/watch?v=o43JaNPlijs&list=PLyeUudrNoHk5981V3gOJLdmgFc4M7cfX-&index=3>

Date: 2-4. May 2023

AP: 7, 2, 6

Location: Fichtenwalde

Occasion: Practical workshop with Nature Foundation David

Participants: Held, around 50 participants from various foundations with fire problems: Federal Forestry; Havel-Oder-Spree Federal Forestry Enterprise; European Forest Institute; Heinz Sielmann Foundation; Central Thuringia Landscape Conservation

Association; Central Thuringia eV; NABU National Nature Heritage Foundation; Lower Oder Valley National Park - Administration; Harz National Park Administration; Saxon Switzerland National Park Guard; David Nature Foundation; PYRECO; Redell Arbeitssicherheit GmbH; Beelitz City Forest District Manager; Sachsenforst State Enterprise; Brandenburg Natural Landscapes Foundation - The Wilderness Foundation; Brandenburg Nature Conservation Fund; Mecklenburg-Vorpommern Foundation for the Environment and Nature Conservation; SUNK; TU Dresden, Institute of Silviculture and Forest Protection.

Short description: Workshop and training "Fire watch and extinguishing"

Date: April 29, 2023

AP: 7, 3

Location: Potsdam Mittelmark

Occasion: Handover of 2 skid units

Persons involved: Held, Fire Service Technical Center Beelitz-Heilstätten

Short description: Thanks to trustful networking, the WKR project was able to build two so-called *skid units*

from the manufacturer Vallfire, developed for fighting forest fires on 4x4 pick-up vehicles, for the Potsdam-Mittelmark district. Both modules were handed over to KBM Jens Heinze at the Beelitz-Heilstätten Fire Service Technical Center at the beginning of May. From there, they will be deployed to various fire departments in the district. The vehicles will also be equipped with hand tools, backpack sprayers, and backpack leaf blowers, thus forming a powerful operational tool in conjunction with other resources. The Potsdam-Mittelmark district is already experiencing numerous fires and is working with KBM Heinze on the future-oriented WKR project to address the new challenges ahead. We are pleased to be able to support these efforts through effective networking.

Date: April 24, 2023

AP: 7, 6, 5

Location: Wildau, Brandenburg

Occasion: Demonstration of UXO fire management technology

Persons involved: Held and Pronto; Spree-Neisse; Potsdam-Mittelmark; Dahme-Spreewald; Teltow-Fläming; Elbe-Elster; Berlin Fire Department; Explosive Ordnance Disposal Service of the State of Brandenburg; Zeithain Fire Department; SKV; Ministry of the Interior and for Local Affairs of the State of Brandenburg; Brandenburg State Forestry Authority; State School and Technical Facility for Fire and Disaster Protection of the State

Brandenburg; Federal Forest.

Short description https://www.digitalmagazin.de/marken/forsttechnik/hauptheft/2023-6/forest-protection/030_remote-controlled Demonstration of the “UXO Fire Management Toolbox” in Brandenburg with the company DokIng from Croatia

Date: April 20, 2023

AP: 7, 2

Location: FBZ Karlsruhe

Occasion: “Forest Edge Training”

Participants: Held, 20 participants from forestry administration and Forst BW from all over Baden-Württemberg

Date: 03 March 2023

AP: 7

Location: Kreuzau NRW

Occasion: Controlled burning, Drover Heide

Participants: L. Pronto; M. Schlossmacher; Nature Foundation; Forest Fire Team; Farmers

Brief description: Approximately 14 hectares of Drover Heath were burned for nature conservation purposes. This approach rejuvenates and revitalizes over-aged stands, providing new habitat for rare animal and plant species. Controlled burning, combined with grazing by cattle and goats, has been successfully used in Drover Heath since 2007.

Date: March 1, 2023

AP: 7

Location: Online

Occasion: “Forest Europe” workshop on forest fires in Ukraine

Participants: Forest Europe, Held, Pronto, Ukraine

Short description: WKR as an example of capacity building and EoE for Ukraine

Date: 22-24 September 2022

AP: 7, 1, 2, 6,

Location: Gartow, Lower Saxony

Occasion: Gartow Demo Days for FNR

Participants: L. Pronto, A. Held, M. Schloßmacher, Ms. Plothe (FNR); dozens of stakeholders participated, including technology providers, forestry authorities, 150 emergency personnel from five districts, the press, and civil protection.

Short description: The WKR project organized an action day with training and presentations on fire management at the demonstration site in Gartow, Lower Saxony. The open day aimed to introduce preventive fire protection measures such as fuel reduction, buffer zones, and control lines to the FNR, WKR partners, and the media. A Dräger simulation training device was also presented. As part of a tactical demonstration, various ignition patterns and suppression techniques were applied on 1 hectare of heathland, and an intense “forest fire” was simulated to demonstrate the effectiveness of the counterfire operation.

Follow up: 3 videos for AP6, press, FNR public relations

Date: August 27, 2022

AP: 7

Location: Chiemgau Alps (Geigelstein)

Occasion: Swathing and training of alpine farmers in fire extinguishing work

People involved: L. Pronto

Date: 6. August 2022

AP: 7

Location: Chiemgau Alps (Geigelstein)

Occasion: Swathing and training of alpine farmers in fire extinguishing work

People involved: L. Pronto

Date: July 19, 2022

AP: 7

Location: Munich

Occasion: Workshop as Interforst Side Event

Participants: L. Pronto, A. Held; FVA, PCF, OTH, THW, BOSCH, Skyseed, LFS-Regensburg, Forst

BW, Fraunhofer, Prepared International, Dräger, Bavarian Insurance Chamber, Ministry of Food, Rural Areas and Consumer Protection Baden-Württemberg, Board of Trustees for Forestry and Forestry Technology, OroraTech, Wahlers Forsttechnik

Short description: See: <https://www.waldbrand-klima-resilienz.com/post/future-challenges-of-forest-fires-in-germany-workshop-results-of-june-17-2022>

Date: July 16, 2022

AP: 7

Location: Chiemgau Alps (Geigelstein)

Occasion: Swathing and training of alpine farmers in fire extinguishing work

People involved: L. Pronto

Date: April 4, 2022

AP: 7, 3

Location: Füssen, Bavaria

Occasion: Pilot project meeting

Participants: L. Pronto, OTH, Bavarian Mountain Rescue Service, LFWS Würzburg

Short description: Mountain rescue, fire brigade, forestry exercise and either WKR demonstration or follow-up project – Topics: rough terrain, tourism, cultural heritage (royal palaces)

Date: 28-30 March 2022

AP: 7

Location: Jüterbog, Brandenburg

Occasion: Inspection and establishment of demonstration areas, training

Participants: A. Held, L. Pronto, NLB

Date: 25-27 March 2022

AP: 7, 1, 2, 3

Location: Goslar, Saxony-Anhalt

Occasion: Training "Forest Fire Advisor"

Participants: Goslar City Forest, Regensburg Forest Service, Bavarian Mountain Rescue Service, Erlangen Fire Department, Baden-Württemberg State Forestry Administration, Baden-Württemberg Forestry AöR, Lower Saxony State Forestry Service, Lower Saxony Fire Service Academy, Brandenburg State Fire Service School

Brief description: In a two-day training session, participants from various federal states and institutions, including forestry and fire services, practiced handling forest fires in difficult terrain in the Harz Mountains. The region faces an increased risk of forest fires due to the widespread dieback of spruce forests. The goal of the exercise was to improve the participants' skills in situation assessment, fire analysis, basic tactical forest fire forecasting, and communication with local operations command.

Follow-up: The exercise promoted interdisciplinary collaboration and particularly highlighted the added value of a forest fire expert advisor to optimize procedures and align understanding and approaches between forestry and firefighting personnel. It also provided an opportunity to strengthen the "forest fire network" through informal discussions. Special thanks go to the forest fire team for the professional planning and coordination of the exercise.

Date: 04 March 2022

AP: 7

Location: Kreuzau NRW

Anlass: Prescribed Burning, Drover Heide

Participants: L. Pronto; M. Schlossmacher; Euro-Waldbrand; Nature Foundation; Nigedden Fire Department (M. Pütz); Forest Fire Team

Short description: Around 12 hectares of Drover Heath were burned for nature conservation purposes. This rejuvenates and revitalizes overgrown stands. Rare animal and plant species are thus provided with new habitat. Controlled burning combined with grazing by cattle and goats has been successfully used in Drover Heath since 2007. This is certainly one of the rare long-term fire-grazing projects in Germany! Without these maintenance measures, the heathland would become overgrown with grass and bushes – even leading to the development of forests.

Date: 08-09 February 2022

AP: 7, 3, 1

Location: IFV (Fire Academy / Institute for Safety) Arnhem, Netherlands

Occasion: Training top national forest fire advisors (NL)

Participants: IFV (Brandweeracademie / Institute for Safety; Technosylva; Wageningen University; Belgian Fire Service

Short description: Training of the 9 national tactical advisors for wildfires and the national wildfire experts with WKR training concept on fire behavior, fire weather, operational safety, and firefighting tactics; participation of Belgian fire brigade commanders for a table-top exercise on cross-border wildfires between Belgium and the Netherlands. The WKR training materials were translated into English to train the fire brigade leaders. One of the main objectives was to create a standard and common understanding for cross-border incidents between the Netherlands and Germany, such as the incidents in 2018 and the large wildfire in 2020 in De Meinweg National Park, which spread to Germany and required the cooperation of German and Dutch firefighters.

Follow-up: Discussion on adapting the WKR training modules for the national level

Date: December 1, 2021

AP: 7

Ort: Gartow

Occasion: Working meeting on the Gartow demo area

Participants: L. Pronto A. Held

Date: November 14, 2021

AP: 7, 1

Location: Schmidt/Nideggen (Eifel National Park), NRW

Occasion: Course: WKR training forest edge control

Participants: L. Pronto; 25 firefighters from the Schmidt, Nideggen and Hürtgenwald fire departments

Short description: A one-day seminar on forest firefighting, safety and risk management at the scene, tactical use of fire

Date: 07-12 November 2021

AP: 7, 2

Location: Belson-Bergen, Lower Saxony

Occasion: Course: Wildfire Analysis

Participants: L. Pronto; Pau Costa Foundation; GRAF (Catalonia Bombers); @fire; Federal Fire Service of Lower Saxony; LFS Saxony; LFS Brandenburg

Short description: Support WKR partner PCF in training on forest fire analysis

Date: 02-04 November 2021

AP: 7, 1-2

Location: Luckenwalde /Jüterbog, Brandenburg

Occasion: Part 2 Forest Fire Training

Participants: A. Held, L. Pronto; T. Gausche, Tetow-Flaeming District; EuroWaldbrand; Brandenburg Natural Landscapes Foundation; Zeddenick Fire Department; among others.

Short description: Practical training part: Focus on the use of fire as a tactical measure on ammunition areas

Date: 09 October 2021

AP: 7

Location: Aschau im Chiemgau, Bavaria (Rosslalm, 1600m)

Occasion: Sludge harvesting with mountain farmers; practical training: mop-up

Participants: Local mountain farmers of the region; L. Pronto

Short description: In times of climate change, established/cultural practices such as clearing can have unexpected/unintended consequences. This training emphasized the importance of follow-up firefighting to prevent forest fires in the Alpine region, which have been increasing in recent years due to clearing.

Date: 06-07. July 2021

AP: 7, 6, 3, 8

Location: Bruchsal / Hardtwald

Occasion: Network meeting 2021, topic: forest fire management

Participants: A. Held, L. Pronto; FVA, WSL (CH); BaySF, Sachsenforst; KVK of the Bundeswehr in Rhineland-Palatinate; Euro-Forest Fire; Institute of the Fire Service of North Rhine-Westphalia; Rhein-Neckar District Office; TH Cologne; ForstBW; Thuringian Ministry of the Interior; Thuringian State Fire Service and Disaster Management School (TLFKS); Rhein-Neckar District Office; State Forestry Authority of Rhineland-Palatinate; Lower Saxony Chamber of Agriculture, PRIMA Project; State Forestry Administration of Baden-Württemberg; German Forest Insurance Office; KLT; MLR; THW

Short description: Model regions approach BaWü; Model region Hardtwald and Ortenau; Presentation of the Project WKR and the Hardtwald demo areas

Excursion: Preventive forest fire management (presentation of the possibilities of preventive forest fire management)

Network activities in forest risk and crisis management: developments in Forest Europe Process; status of German initiative & Stihl application; contributions from network partners

Possibilities of mobile first attack: Presentation of mobile devices and tools for fire prevention and first attack in the event of forest fires

Date: 05. July 2021

AP: 7, 3, 8

Location: Online

Occasion: Steering committee meeting Model Region BW

Participants: A. Held, L. Pronto; FVA, ForestBW, Rhein-Neckar-Kreis; Karlsruhe District Office; MLR

Brief description: A multi-stage approach has been agreed upon, which has already proven successful in another model region (see presentation). Thematic focuses will be on the challenges posed by forest fires from the perspective of the respective members and their organizations, identifying potential synergies, information and communication needs among the participants, and determining the need for specialist expertise on specific topics.

Date: June 23-27, 2021

AP: 7

Location: Gartow (demo areas of Bernstorff'sche Betriebe)

Reason: Work on the demo area

Participants: Alex, Rosa, Maria (WKR) plus about 10 people from the forest fire team, ForestFireWatch and local fire brigade

Short description: Working with a mulching machine and hand tools, creating protective strips (blog post: <https://www.waldbrand-klima-resilienz.com/post/waldbrandvorsorge-mit-der-feuerwehr-arbeiten-an-der-demo-fl%C3%A4che-bei-gartow-gehen-weiter>)

Date: April 22, 2021

AP: 7, 6

Location: online

Occasion: David Nature Foundation "Forest fires on natural heritage and wilderness areas: prevention, control and aftercare"

Participants: Alex, Maria

Short description: Alex, keynote speech

Follow up: 2x practical workshop and 1x online workshop

Date: April 15, 2021

AP: 7

Location: Online

Occasion: NLB practical training

Participants: Alex, Lindon, Andreas Schulze, NLB staff

Short description: Theoretical training on safety and basics in WBB for NLB employees

Date: April 12, 2021

AP: 7

Location: "Altes Lager", Jüterbog / Luckenwalde

Occasion: NLB practical training

Participants: Alex, Andreas Schulze, Nico Semsch with 10 people (SKV FFW Zehdenick), EuroWaldbrand, FFW Teltow-Fläming with 10 people

Short description: Practical training in indirect firefighting

Follow up: Online-Theorie

Date: February 16, 2021

AP: 7, 8, 3

Location: Hardtwald, Bruchsal

Occasion: Discussion and brainstorming of the "Model Region Forest Fire in Baden-Württemberg"

Participants: Pronto, Held, FVA, Forst BW

Short description: Brainstorming with the participating forestry departments on the topic of establishing a model region "forest fire" in BaWü

Follow up: Concept paper by April 2021

Date: 15-17 February 2021

AP: 7

Location: Hartdswald, Bruchsal

Reason: Setting up the demonstration areas for the WKR project **People**

involved: Pronto, Held, Forst BW **Short description**

Reduction of fire load along a symbolic skid road, change in the stand structure, plus thinning, pruning, reduction of fire load in 10-year-old pine nature conservation area.

Planning of a "rotating fire prevention with nature conservation areas (here nightjar)

Follow up: Demo area set up in February 2021.

Date: October 9-11, 2020 **AP:** 7, 1-3

Location:

Munster **Occasion:**

Fire camp with fire team **Participants :** Alex, Lindon,

Rosa, Maria (WKR) + 35 people from the forest fire team, Forest Fire Watch, Forestry Office and private forest **Short description:** Training weekend for practical training in controlled burning at the Munster

(Örtze) military training area

Follow up: recap: <https://www.waldbrand-klima-resilienz.com/post/leben-in-der-lage-oder-feuereinsatz-zur-vegetationsbrandbek%C3%A4mpfung>

Date: July 20, 2020 **AP:** 7,

8 Location:

Gartow, Lower Saxony **Occasion:**

Discussion and planning of the selection of demonstration areas for the WKR project **Participants:** Held, Dr.

Herrmann (FFW), and Uli von Mirbach **Brief description :** Presentation of the WKR to

forest owner Graf Berstorff, exchange of ideas, and presentation of demonstration areas. Discussion of the selection criteria.

Follow up: Demo area setup fall 2021

Date: July 16, 2020 **AP:** 7,

6 Location:

Kulmbach

Occasion: Discussion and planning of a prototype fire crawler for forest fire operations **Participants:** Held,

KLP **Short description** Presentation of

WKR at KLP, discussion of the requirements of a remote-controlled mulching crawler for fire management **Follow up:** Prototype developed and in use

Date: July 14-14, 2020 **AP:** 7,

8 Location:

Luckenwalde **Occasion:**

Discussion and planning of the selection of demo areas for the WKr project **People involved:** Pronto, Held, Schuck, TF district, NLB Foundation, TF fire department **Short description** Presentation of WKr to TF district, exchange of ideas and presentation of demo areas.

Discussion of the selection criteria.

Follow up: Training of local partners in April 2021_: Indirect firefighting

Date: July 9, 2020 **AP:** 7,

8 Location:

Muna feucht, Forstborteieb Nuremberg **Reason:**

Discussion and planning Selection of the demo areas for the WKr project **People involved:** Held, BaySf,

Project Advisory Board Kraus D **Short description** Presentation of WKr

at BaySF Exchange of ideas and presentation of demo areas.

Discussion of the selection criteria.

Follow up: Areas are still being cleared of munitions.

Date: July 7, 2020 **AP:** 7,

8 Location:

Hartdswald, Bruchsal **Occasion:**

Discussion and planning of the selection of demonstration areas for the WKR project **Participants:** Pronto,

Held, FVA, Forst BW **Brief description:** Presentation of the WKR

to the Forst BW district of Hartdswald, exchange of ideas, and presentation of demonstration areas. Discussion of the selection criteria.

Follow up: Demo area set up in February 2021.

WP8 Project Coordination and Administration

The WKR project was coordinated and administered by both partners, EFI and FVA. In 2020 and 2021, monthly working meetings (more frequently during the pandemic) were held to plan project activities in the context of the pandemic and the availability of relevant partners. By 2022 and 2023, workflows had settled down, and working meetings were held as needed.

The project was accompanied by an external project advisory board from forestry, fire service and research:

- **Stefan Doerr**, Editor-in-Chief Int. J. Wildland Fire, Professor at Department of Geography, College of Science, Swansea University
- **Thomas Egelhaaf**, State Fire Director of Baden-Württemberg and State Fire Service School Baden-Württemberg
- **Hans von der Goltz**, Federal Chairman of the Working Group for Natural Forest Management ANW Germany
- **Daniel Kraus**, Management Service Bavarian State Forests, Forestry Manager at the University Forestry Office Sailershausen (University of Würzburg)
- **Peter Moore**, Forestry Officer Fire Management and Disaster Risk Reduction, FAO
- **Birgit Süßner**, Fire Chief of the City of Erlangen; Lecturer for forest and vegetation fires in the career training course for advanced technical fire service in Bavaria and the State Fire Service School in Regensburg
- **Diethelm Witthoff**, National Forest Fire Protection Officer, Ministry of the Interior and Local Government of the State of Brandenburg (ret.)

The Project Advisory Board was regularly informed about the planned activities of the project. Individual members of the Project Advisory Board also met at numerous events and conferences, and these meetings were also used for coordination.

WP8-relevant activities:

Date: June 20, 2023

AP: 8

Location: Online

Occasion: FNR status seminar "Forest fire prevention and management"

Participants: L. Pronto, A. Held

Date: 28 September 2021

AP: 8, 4

Location: Online

Occasion: EFI strategic talks with UN-FAO

Meaning Characters: A. Held, L. Pronto; P. Moore/FAO; EFI

Date: August 11, 2021

AP: 8, 3

Location: Online

Occasion: Exchange on the topic of public relations with THOR and Brennpunkt Waldbrand

Participants: M. Schloßmacher

Short description: -

Maintaining contact with journalists

- Dissemination strategies for publishing project results

- Joint public relations between the projects

- Reaching different target groups, ideas and implementation

- Concept design (Are there written concepts for public relations in the projects?)

Follow up: Further meetings planned, input from WKR desired, interview with Prof. Müller for interview series

Date: July 20, 2021

AP: 8, 3

Location: Online

Occasion: Coordination meeting between Project WKR and THOR

Participants: A. Held, L. Pronto, M. Schloßmacher; THOR Project

Short description: Discussion of project synergies and prospects for cooperation in specific areas - in particular training and guidance for munitions-contaminated areas.

Date: June 15, 2021 AP: 8,

7 Location:

Online

Occasion: Planning meeting

Participants: WKR, forest fire team, Forest Fire Watch **Short description:**

Coordination of logistics and plan Gartow plus Lübtheen Minister visit to THOR project **Follow up:** Demo

Date: June 10, 2021 AP: 8,

7, 3 Location:

Online

Occasion: WG: Planning Network Meeting

Participants: WKR, FVA and Forst BW **Short description:**

Meeting to plan network meetings, trainings and demonstrations (model region BW)

Follow up: Further planning meetings

Date: April 7, 2021 AP: 8

Location:

Online **Event:** FNR Status Seminar

Participants: Lindon, Alex, Maria **Brief description:**

FNR meeting of funded forest fire projects. WKR was one of five (?) projects that was invited to present itself. Maria introduced WKR, and Alex and Lindon were present for Q&A.

Follow up: Good feedback, Alex invited DWD guys to Wildfire Conference.

Date: March 30, 2021 AP: 8

Location:

Online **Event:** WKR team meeting (FVA and EFI)

Participants: Christoph, Carolin (partially Yvonne), Alex, Lindon, Andreas, Maria **Short description:** Update WKR meeting **Follow up:** WKR meeting May/June? (1 year after kick-off)

Date: February 23, 2021 AP: 8,

3 Location:

Online

Occasion: Exchange with Project THOR

Participants: WKR team, FVA, THOR **Short description of mutual introduction of WKR and THOR.**

Follow up: FNR webinar on all FNR-funded fire projects.

Date: November 24, 2020 AP: 8

Location:

Online **Occasion:** Project Advisory

Board Meeting **Participants:** Project Advisory

Board **Short description:** Presentation of the completed and planned work, discussion of the role of the Advisory Board **Follow up:** repeat

Date: August 2020 AP: 8,

7 Location:

Neuhausen auf den Fildern, **Reason:**

Planning meeting with THW federal school, discussion of THW's role as a supporting partner in the WKR project **People involved:** Hagen Vollrath, A Held

Short description: The content was a plan for FireLine

Explosives, chain saw etc., possibilities for cooperation, which priorities?

Date: July 13, 2020 AP: 8

Location:

Online **Event:** WKR kick-off

Participants: Tobias

Wiepcke, KWF, Thomas Egelhaaf, FWS-BW, Meissner, NLB Foundation, Schulze, NLB Foundation, Tino

Gausche, KBM-Teltow-Flaeming, Sebastian Hanussek, Stihl, D. Maushake, forest fire team, Michael Herrmann, Forest Fire Watch, Mirbach, Gräffiche Bernstorff, Bernd Heinrich, KWF, R. Birkefeld, Energreen Germany, Hagen Vollrath, THW, A. Lyncker, FNR, M. Plothe, FNR (+WKR team plus project advisory board)

Short description: Kick-off/official project launch (virtual due to the circumstances) with the key partners for the project. Presentation of the project content, the team, and further plans. There was an opportunity to get to know each other (virtually) through profiles completed by the partners in advance, and participants from forestry, industry, municipalities, and fire departments exchanged experiences, knowledge, and suggestions directly. PowerPoint presentation in the WKR folder.

Follow up: There is great interest in a continuous exchange between those involved. Next meeting of this kind expected at the end of the year.

2. Utilization

- the expected benefit, in particular the usability of the result in the sense of updated utilization plan
- Scientific and/or technical prospects of success after the end of the project or prospects for a possible necessary next phase or the next innovative steps for the successful implementation of the results
- economic prospects of success

Work packages (WP)	Recycling
AP 1 Training modules for Fire departments	The modules were presented at all WKR project events. The feedback was consistently positive. The acceptance and adoption of the WKR materials by the National Training and Tactics Working Group is a tremendous testament to the quality and acceptance of the WKR modules. Several other institutions, state fire service schools, and instructors are also using our training materials, and we receive numerous requests for wider use. At the end of the project, we also received requests from Belgium, the Netherlands, Poland, the Czech Republic, Lebanon, and Luxembourg for translations of the modules into their respective national languages. Unfortunately, we are no longer able to support this process. Since the WKR modules have been used at the state fire service academies and by the UAG, inquiries from practitioners have multiplied. In 2024 alone, i.e., after the actual project duration, we received around 35 direct inquiries from fire departments regarding the use of the modules. It can therefore be assumed that WKR products will remain in use for many years to come.
AP 2 Training modules for forestry AP2 Training modules for preventive Fire protection through silviculture	The manual on silvicultural forest fire prevention was developed in close collaboration with the Baden-Württemberg Forest Fire Protection Agency (FVA) and has been distributed in training courses for forestry enterprises and administrations in the federal states of Lower Saxony, Brandenburg, Saxony, Bavaria, and Baden-Württemberg. The feedback has been consistently positive. Forestry training centers such as the FBZ Karlsruhe, Mattenhof (both Forst BW), the Brandenburg Forestry School and the Ostheide Forest Education Center, communities such as Fichtenwalde-Beelitz, the University of Freiburg and the Klimaarena Mannheim are users of the silvicultural modules. It can therefore be assumed that WKR products will remain in use for many years to come.
AP 3 Expertenaustausch (Exchange of Experts - EoE) and networking	Through the expert exchanges, many participants from Germany were able to gain international experience in the field of forest fires for the first time and often also their first practical experience. The contacts and networks that have been created through the EoEs continue to exist and continue to bear fruit. The independent relationships and networks established through our EoEs continue to be highly effective in increasing capacity within federal states. For example, our EoEs have led to direct relationships between state fire schools and professional agencies and NGOs in France, the Netherlands, Poland, Spain, Portugal, Croatia, Italy, South Africa, the UK, the US, and Canada. Competencies that were built up in and through the EoEs now flow directly into the emerging Forest fire competence centers (Brandenburg, Saxony, Baden-

	Württemberg, Bavaria). A special application is the new concept for forest fire management on UXO areas developed by the state of Brandenburg: The EoE to Croatia also addressed the question of new approaches to fire management on munitions-contaminated areas. The experience gained in Croatia, new technology, and exchanges with fire managers fighting fires on UXO areas in Croatia convinced the state of Brandenburg to pursue new approaches. As a result, Brandenburg has invested €4 million and is developing innovative methods for fires on UXO areas using new technology. As a further utilization, the state of Saxony is now in the process of deciding whether to introduce this technology and concept. It can therefore be assumed that the results, networks, structures and innovations created by WP3 will remain in use for many years to come.
WP 4 Development of the framework and Core contents of a general Forest fire strategy	A national forest fire strategy, or the process of developing one, does not exist. However, WKR has been able to influence strategy development in at least the individual federal states; the planned Forest Fire Competence Center for Brandenburg, Baden-Württemberg, and Saxony is one example. WKR was also able to present the necessary key points of a strategy to the German Bundestag. Furthermore, WKR was able to support this process in numerous expert committees and panel discussions, and in cooperation with associations such as KWF, ANW, DFWR, SDW, etc. A national forest fire strategy would require consensus and the will of the relevant ministries; this is currently not apparent. WKR has also supported the international coordination of the core contents of a forest fire strategy, participated in strategy development at the EU and UN levels, and thus laid the foundation for the development of a German forest fire strategy in coordination with international processes. not only provides the absolutely necessary content of a strategy, but also enables Germany's potential involvement in international "Fire Policy and Governance" processes. It can therefore be assumed that the WKR products will continue to be useful in the development of a national strategy even after the project has ended.
WP 5 Technical recommendations for policymakers and decision-makers	The WKR's expert recommendations, the countless media contributions related to them, and the awareness-raising efforts through seminars and conferences are increasingly finding their way into political decision-making. With the end of the WKR project, a driving force in the political landscape is naturally lost, keeping the issue of forest fires firmly in the political consciousness. Associations such as the German Fire Brigade Association, the UAG Training and Tactics, the German Forest Protection Association, the Working Group for Natural Forest Management ANW and the Board of Trustees for Forest Management KWF have adopted the WKR recommendations and will continue to engage in political lobbying for them. It can therefore be assumed that WKR products will remain in use for many years to come.

WP 6 Awareness work and Raising awareness	The website served as a central source of information for project activities, relevant publications and network news throughout the project. The channels and reach of the European Forest Institute, as well as the WKR network, were also continuously used to raise awareness and promote project content. Raising awareness about WKR activities also had an international impact, which in part secured more participants and expanded the network for both project activities and multipliers. For example, WKR project activities and results were presented to the REGI Committee of the European Parliament, at the high-level 4th Nicosia Risk Forum in Cyprus, at the Fire and Climate Conference (International Association of Wildland Fire) in California, at the Northwest European Round Table on Fire Management (Netherlands), and at the Bonn Network for International Disaster Risk Reduction. -Media work -Forest fire education In the last months of the project, in cooperation with various state-owned companies, a nationwide "Forest fire danger sign" is being worked on, which is in the final stages of completion and can be presented soon. Overall, the input and practical impact of WP6 can be considered extraordinary, especially in comparison with other or previous forest fire projects.
AP 7 demonstration areas and Trainings, Workshops	Work on the demonstration areas has continued, among others in North Rhine-Westphalia, Brandenburg, Lower Saxony, Bavaria and Baden-Württemberg. The FNR Media Day 2022, a public exercise at the WKR demonstration area in Lower Saxony, was a complete success. In the "Chiemgau Alps" demonstration region, several actions and training activities on the subject of silting and post-fire extinguishing work were carried out together with mountain farmers in 2022. In March 2022, a large-scale "Forest Fire Advisor" exercise was carried out with the WKR project partner Forest Fire Team In August 2022, WKR organized a forest fire workshop as a side event at Interforst in Munich with the EU project FIRE-IN (DG-Home). In addition, WKR participated in or organized several other workshops and trainings together with partner organizations and project partners. The landowners, associations such as ANW and SDW, state fire service schools and forestry training centers will continue to use the demonstration areas. It can therefore be assumed that the WKR products from AP7 will remain in use for many years to come. Nevertheless, as in all APs, the driving force, the source of ideas, motivator and inspiration of the WKR project is now gone.
WP8 Project coordination and Administration	Project coordination and administration were continued on an ongoing basis, always in close cooperation with the partner FVA. Regular formal and informal exchange with the external

	The project advisory board ensured not only the quality but also the dissemination of the WKR results.
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3. Findings from third parties

- progress in the field that has become known to the ZE during the implementation of the project of the project at other locations

During the project period, the WKR project was able to actively support other bodies:

Model region for forest fires in Baden-Württemberg

Development of forest fire education with the WPZ Ostheide (Forest Education Center)

Concept development and technology procurement for fire on munitions-contaminated areas of the country Brandenburg

Cooperation and consulting with industry, for example the cooperation with BOSCH in the field of ultra-early fire detection.

Through exchanges with remote sensing companies, such as Ororatech, but also start-up companies from the international sector, WKR was always able to promptly evaluate the findings of third parties and incorporate them into the project work.

4. Publications

- the publication of the results that has taken place or is planned

All publications within the WKR project were promoted through various channels. These included the WKR project's reach (website, networks of WKR forest fire experts, the development of new networks during the course of the project), but also by leveraging the international reach of the EFI, primarily through its website and various social media channels. Publications were published in trade publications such as AFZ and Forst und Technik, as well as in print, radio, and television. A 45-minute **documentary** is currently in development with SWR and will be broadcast in mid-2024.

For details see <https://www.ardmediathek.de/video/swr-story/risiko-waldbrand-wie-gut-sind-unsere-feuerwehren-vorbereitung/swr/Y3JpZDovL3N3ci5kZS9hZXqvbzlwODg1ODU#:-:text=Risiko%20Waldbrand%20%2D%20Wie%20well%20are%20our%20fire%20departments%20prepared%3F-01.08.2024%20%E2%88%99&text=Waldbr%C3%A4nde%20zerst%C3%B6rten%202022%20in%20Deutschland.war%20betroffen%20%2D%20trotz%20starker%20Niederschl%C3%A4ge.>

Total during the project period:

Interviews: Print/online (96), Radio (40), TV (47.), Podcasts (5), Feature Articles (3) WKR-Blog posts (41), WKR newsletters (2), WKR videos (13), documentary film (1)

Media work is ongoing and many of the interviews mentioned above have been printed in several newspapers or appeared in several languages.

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